Listening to History Rhymes: The Geometry of Stock Return Trajectories

Abstract

We organize 1.3 million standardized 60-day U.S. equity return trajectories from 1971 to 2025 through exhaustive pairwise similarity into a geometric space. We propose a random-walk-adjusted *Pattern Distance Ratio* (PDR) to capture recurring structures without specifying their forms in advance, unlike conventional price-pattern research. Recurrence is genuine and persistent: PDR's year-to-year correlation is 0.308, versus 0.011 for a placebo. Location in the geometric space contains information about subsequent returns: it predicts direction better than chance, while a strategy based on the similarity representation earns gross annual alphas of 6.0% to 9.2% and positive equal-weighted alphas in twelve international markets.

Keywords: return predictability, trajectory space, matrix profile, time series analysis, machine learning, technical analysis.

History never repeats itself, but it does often rhyme.

— Mark Twain

1 Introduction

Traditional finance theory, notably the efficient market and random walk hypotheses, has long provided a foundational framework for understanding stock price dynamics. At the same time, a large literature has documented numerous market anomalies (e.g., Keim, 1983; Ang et al., 2006; Banz, 1981; Jegadeesh and Titman, 1993) and related return regularities to human behavior (e.g., Stambaugh et al., 2012; Odean, 1998; Shefrin and Statman, 1985), market imperfections (e.g., Duffie, 2010), and predefined statistical rules such as those used in technical analysis (e.g., Han et al., 2013; Zhu and Zhou, 2009; Kwon and Kish, 2002). Yet our understanding of the mechanisms driving markets remains necessarily incomplete, given the large number of participants, the complexity of market mechanisms, and the interactions among the many forces that jointly determine prices. Empirical work therefore necessarily makes choices about which aspects of return trajectories to retain and examine.

More broadly, empirical studies of return dynamics typically begin by compressing a stock's return trajectory into a set of researcher-chosen variables. To make this distinction explicit, let $R_{i,t} = (r_{i,t-L+1}, \ldots, r_{i,t})$ denote a recent return trajectory of length L. Conventional approaches generally construct a lower-dimensional representation $X_{i,t} = \phi(R_{i,t})$, where ϕ may capture cumulative return, volatility, reversal, recent extremes, trends, or other predetermined features of the path, and then study subsequent returns conditional on $X_{i,t}$. Such compression facilitates both economic interpretation and computational tractability, but it necessarily requires the researcher to decide ex ante which aspects of the return trajectory to retain. We instead retain the return trajectory $R_{i,t}$ itself as the empirical object. In our baseline analysis, we use $L = 60$ trading days, while examining alternative window lengths for robustness. As modern computing technology increasingly relaxes the computational constraint, retaining the trajectory itself becomes empirically feasible. Doing so shifts our attention to a different empirical problem: how to organize

return trajectories and characterize their relationships to one another.

The *Matrix Profile* methodology (Yeh et al., 2016), originally developed for motif discovery in time-series data (Lin et al., 2002), provides a natural foundation for addressing this problem. By comparing time-series segments through z-normalized Euclidean distance, the Matrix Profile supplies a geometric structure for a collection of otherwise individual trajectories: each trajectory can be viewed as a point in a high-dimensional metric space, with distances defining similarity and nearest neighbors defining local neighborhoods. This geometric representation gives us both a way to organize return trajectories and a set of native empirical objects: distance, neighborhood, recurrence, and location. Location here is relational rather than coordinate-based: it is characterized by a trajectory's distances, or similarities, to other trajectories in the space. We accordingly construct a geometric space—which we refer to as trajectory space—from approximately 1.3 million standardized 60-day rolling return trajectories across stocks and time spanning 1971–2025.

A defining feature of our approach is exhaustive comparison: each return trajectory is compared with all other trajectories within trajectory space. This exhaustiveness allows the search to remain assumption-light with respect to the forms that return patterns may take. We do not require return paths to repeat exactly, nor do we specify in advance which shapes, signals, or chart formations should matter. Instead, similarity is defined through distance in trajectory space, allowing recurring structures to emerge from the data. In this sense, history rhymes without repeating, and the exhaustive search allows us to listen for those rhymes without insisting in advance on what we should hear.

Within finance, the most direct antecedent of our approach is Lo et al. (2000), who put the reading of price patterns on a rigorous footing. They use nonparametric kernel smoothing to automate the identification of familiar chart formations and then examine whether these formations contain information about subsequent returns. Their approach demonstrates that visual patterns can be transformed into algorithmic and statistically testable objects rather than left to chartist judgment. Our approach is complementary. Whereas Lo et al. (2000) organize price trajectories through a predefined set of economically familiar patterns—such as head-

and-shoulders, broadening tops, and triangles—we organize return trajectories directly through pairwise similarity in trajectory space. The former provides interpretable pattern categories; the latter allows recurring structures to emerge without requiring that they first be assigned to named formations, shifting attention more broadly to the geometry of return trajectories as an empirical object in its own right.

We use this geometric perspective to address two broad questions. First, does trajectory space itself exhibit persistent structure? The structure we examine is the recurrence of similar trajectories across the space. We view a trajectory as exhibiting greater recurrence when it has more closely similar occurrences in the sample, with similarity measured by distance in trajectory space. We examine whether this recurrent structure persists out of sample by comparing the recurrence identified in one period with that observed in a subsequent period. Persistence therefore means that regions of trajectory space exhibiting greater recurrence in sample continue to exhibit greater recurrence out of sample.

To measure recurrence, we follow the nearest-neighbor principle underlying the Matrix Profile and construct the Pattern Distance Ratio (PDR), which compares a trajectory's nearest-neighbor distances in the observed data with those in a matched random-walk benchmark constructed by reshuffling returns across stocks within each calendar year. Benchmarking is essential because nearest-neighbor distances are not homogeneous across trajectory space even under the null: some regions are naturally more densely populated than others. By comparing each observed neighborhood with its corresponding random-walk neighborhood, the PDR accounts for this locational heterogeneity and provides a continuous measure of recurrence relative to the null.

Our second broad question is whether location within trajectory space contains information about subsequent returns. We approach this question first at the neighborhood level, asking whether similar historical trajectories tend to be followed by systematically related outcomes, and then at the level of the full geometric representation, using a trajectory's similarities to historical trajectories to predict future returns and form portfolios. The distinction is deliberate. The first exercise asks whether location in the space contains persistent directional information; the second asks whether that information is sufficiently systematic to matter economically. Together, these

exercises allow us to study the geometric structure of trajectory space before asking how much of that structure can be converted into return prediction.

Our empirical analysis yields several findings. First, trajectory space exhibits genuine and persistent local structure, manifested in the recurrence of similar trajectories. Put differently, regions more densely populated by similar trajectories than the random-walk benchmark implies in sample tend to remain so out of sample. Under annual comparison blocks, the correlation between in-sample and out-of-sample PDR is 0.308 at a one-year distance and remains 0.270 at a three-year distance. Lengthening the comparison windows to two, three, and five years raises the correlation to 0.425, 0.510, and 0.607, respectively. A longer window gives each PDR a larger comparison sample, and the resulting increase in persistence is consistent with our framework. In contrast, a random-walk placebo constructed by reshuffling returns across stocks within each year produces a persistence correlation of only 0.011, essentially zero. The persistent local organization observed in actual return trajectories therefore largely disappears under the random-walk null.

Transition matrices provide a complementary nonparametric view of the structural stability of trajectory space. Trajectories in the extreme PDR quintiles—both the most recurrent (Q1) and the most unique (Q5)—show the strongest tendency to remain in the same quintile over time, with retention probabilities of 26.7% to 34.8% in Q1 and 39.5% to 57.1% in Q5 across window lengths, versus a null benchmark of 20%. More broadly, the largely smooth off-diagonal decay suggests that the organization of trajectory space is itself persistent: trajectories tend to move gradually through neighboring regions rather than shift abruptly across the space. The persistence is therefore not confined to a collection of isolated recurring patterns. Although our subsequent analysis focuses primarily on the highly recurrent region, this broader structure across the PDR distribution suggests that the geometry of trajectory space contains additional regularities that remain to be explored.

We next examine whether this geometry is related to subsequent outcomes. A sign-preservation analysis shows that the direction of a trajectory's neighbor-outcome score holds out of sample more often than chance, and increasingly so as the score becomes stronger: preser-

vation rates rise from about 50.5% for the weakest scores to about 53% for the strongest, against an independence-adjusted null close to 50%. Among trajectories with strong scores, the rate exceeds the null in every PDR quintile. Thus, trajectories occupying similar locations in trajectory space are followed by systematically related subsequent outcomes. This provides evidence that the geometry is not merely persistent: location within it is economically informative.

Finally, we ask how much predictive information can be extracted when we use the geometry of trajectory space more fully. We map each return trajectory to its location in trajectory space—its Gaussian-kernel similarities to a set of landmark trajectories from the previous five years—and predict future returns by ridge regression on this high-dimensional representation, in the spirit of the virtue-of-complexity literature (Kelly et al., 2024). Sorting firms each month into a long–short portfolio by predicted return, while controlling for reversal at the relevant horizons, the strategy earns large gross risk-adjusted alpha—roughly 6.2% to 9.2% per year across the CAPM and Fama–French three- through six-factor models and equal- and value-weighted portfolios—that remains significant in the most recent subperiods. The strategy's high turnover, however, leaves it a break-even one-way trading cost of about 15 to 20 basis points. The economic message is therefore not that exhaustive search uncovers an easily exploitable trading rule. Rather, the portfolio evidence provides a demanding test that information embedded in the geometry of return trajectories is sufficiently systematic to survive aggregation into return forecasts. We further transplant the engine to twelve international equity markets, retaining the U.S. estimation design while re-estimating the model within each market: the equal-weighted long–short alpha is positive in all twelve, and a global country-average portfolio earns a significant return.

Our study relates to a long literature on technical analysis. The idea that the shape of past price movements may contain information about subsequent returns has long attracted investment practitioners, and Menkhoff (2010) documents widespread reliance on technical analysis among fund managers across countries. On the academic side, while early research cast doubt on the records of published market forecasters and mechanical trading rules (e.g., Cowles, 1933, 1944; Fama and Blume, 1966), subsequent studies broadened the systematic investigation of technical patterns across trading rules, markets, and increasingly flexible pattern representations, documenting pre-

dictive structures in historical price movements (e.g., Brock et al., 1992; Bessembinder and Chan, 1995; Lo et al., 2000).

Our study takes this progression toward broader systematic search a step further by changing what is being searched. Conventional technical analysis typically maps the return trajectory into a predefined representation—a trading signal, rule, or recognizable pattern—and asks whether that representation predicts subsequent returns (e.g., Sullivan et al., 1999; Bajgrowicz and Scaillet, 2012; Brock et al., 1992; Neely et al., 2014; Goulding et al., 2024). We instead retain the return trajectory itself and examine the geometric structure of trajectory space.

Bajgrowicz and Scaillet (2012) revisit technical trading rules on the Dow Jones index and show that an investor could not have reliably selected the future best-performing rules ex ante. In our attempt to translate the geometric structure of trajectory space into a trading strategy, we reach a similar economic boundary, with reasonable transaction costs mostly eliminating profitability. Our key insight is that exhaustive search documents persistent recurrence in return trajectories and systematic relationships between similar trajectories and subsequent outcomes. The recurrence survives a block-bootstrap null and reappears in twelve international markets. Both the recurrence and the neighbor-outcome results survive a strict neighbor rule.

Our study also relates to the growing use of machine learning and high-dimensional methods in empirical asset pricing. Gu et al. (2020) demonstrate the predictive power of tree-based and neural-network methods relative to traditional linear models. Leippold et al. (2022) document substantial predictability in Chinese equity markets using machine-learning techniques. Jiang et al. (2023) employ convolutional neural networks to extract predictive information from transformed price-chart images. Murray et al. (2024) feed the past twelve months of cumulative returns to a convolutional network with long short-term memory and find cross-sectional predictability that is distinct from momentum, reversal, and existing technical signals. Chen et al. (2024) develop adversarial deep-learning methods for conditional asset pricing. While flexible machine-learning methods allow increasingly rich mappings from historical information to future returns, our focus is instead on organizing return trajectories and studying the structure induced by their similarities.

Closest in spirit is the contemporaneous study of Chen et al. (2025). We share the broad idea that similarity to historical states can provide a useful empirical organizing principle, but differ in focus and perspective. Chen et al. (2025) construct a news-based representation of the economic state for market-return prediction. We instead make the geometric organization of individual-stock return trajectories the central object of empirical investigation.

For return prediction, we use the full high-dimensional similarity representation in the spirit of Kelly et al. (2024). Nagel (2025) cautions that such high-dimensional kernel forecasts, when trained on short windows, can reduce to recency-weighted momentum; Section 5.1 explains why our design makes this problem unlikely.

Implementing our assumption-light search at scale requires substantial computation. Each trajectory is compared with every trajectory formed in the five years ending in its own, or in the five following years, and every comparison is repeated for the three benchmark panels. Approximately 1.3 million return trajectories therefore require about 1.3 trillion distance calculations.[1] We use GPU-accelerated matrix operations in place of the standard FFT-based Matrix Profile implementation. This computational approach makes it feasible for us to search broadly without reducing the space in advance to a researcher-selected set of features.

2 Data and the Measurement of Recurrence in Trajectory Space

This section describes Stage I of our method (Figure 1): how we measure whether a return trajectory recurs. Stage I addresses the first of the two questions posed in the introduction, whether trajectory space has persistent structure. Stage I also records the later returns of each trajectory's nearest neighbors. Section 4 uses them to ask the second question directly, whether location in trajectory space carries information about subsequent returns, and Stage II (Section 5) answers it with a return forecast and a portfolio. We first describe the data and the return trajectories, then the distance between two trajectories, the random-walk benchmark, and finally

[1] The count sums the in-sample and out-of-sample distance matrices of Section 2.4 over the annual blocks and multiplies by four, once for the real panel and once for each benchmark panel. It covers the baseline 60-day specification; the robustness checks of Section 3.3 repeat parts of it.

the Pattern Distance Ratio (PDR), our measure of recurrence.

2.1 Data

We use daily stock market data from the Center for Research in Security Prices (CRSP), annual accounting data from Compustat, and daily and monthly Fama–French factor returns from Kenneth French's data library, all obtained through Wharton Research Data Services (WRDS). The CRSP data run from January 1970 to December 2025. The daily return sample begins in July 1971, because the benchmark sorts below use each stock's market equity at the end of the previous December.

The sample includes all common stocks (share codes 10 and 11) listed on the NYSE, AMEX, and Nasdaq (exchange codes 1, 2, and 3). Daily returns include delisting returns, following the standard procedure. Market equity is $ME_{j,t} = |P_{j,t}| \times SHROUT_{j,t}$,[2] summed across all share classes of each firm (*permco*). A firm with several share classes enters the sample once, through its class with the largest market equity. Following Fama and French (1993), book equity is $BE = SEQ + TXDITC - PS$, where preferred stock is measured by its redemption, liquidating, or par value, in that order. We require $BE > 0$ and at least two years of Compustat data, to reduce backfill bias. To construct characteristic-adjusted returns, we form 25 benchmark portfolios by sorting stocks independently into size and book-to-market (BM) quintiles, following Fama and French (1993).

Illiquid and very small stocks can produce trajectories whose shapes reflect microstructure noise rather than genuine price dynamics. We therefore exclude a stock-year if the stock's market equity at the end of June is below the 10th percentile of NYSE firms or its closing price at the end of June is below \$5. The size filter removes microcaps that trade too little to produce reliable trajectories, and the price filter removes penny stocks whose returns are dominated by bid–ask bounce. We apply both filters every year. The final sample covers 55 years (1971–2025) and contains about 1.3 million return trajectories. Table 1 traces the sample through these screens

[2] All variables and notation are defined in Table A1 in Online Appendix A.

and describes it by decade.

2.2 Return Trajectories and Matrix Profile Distance Metrics

Our unit of observation is a return trajectory: the daily returns of one stock over a 60-day window. The windows roll monthly and end on *month-end dates*, the dates on which the portfolios of Section 5.1 are formed. For each month-end trading day t, the *formation window* is the 60 trading days up to and including t. Stage I uses these trajectories to measure recurrence. Stage II, described later, asks whether a trajectory's location in trajectory space predicts the stock's return over the *evaluation window*: the 20 trading days immediately after the *formation window*.

Return Trajectories

Trajectory (i, t) is the 60-day return series of stock i that ends on day t:

$$\mathbf{R}_{i,t} = \big(r_{i,t-59},\ r_{i,t-58},\ \ldots,\ r_{i,t}\big), \tag{1}$$

where $r_{i,s}$ is the daily return of stock i on trading day s. A trajectory needs a CRSP record on every day of its *formation window*.[3] A trajectory also needs a record on every day of its *evaluation window*, so a stock that leaves CRSP within those 20 trading days contributes no trajectory. For brevity, we often write a trajectory as $n = (i, t)$, and its stock and date as $i(n)$ and $t(n)$, or simply i and t.

The Matrix Profile Framework and Distance Metric

How should we measure whether two trajectories look alike? We follow the Matrix Profile of Yeh et al. (2016), which extends earlier work on motif discovery (Lin et al., 2002). The Matrix Profile compares all pairs of subsequences in a time series by their z-normalized Euclidean distance. For

[3] Missing daily returns are set to zero before the delisting return is compounded in, following the standard CRSP merge.

two subsequences $\mathbf{x}_i$ and $\mathbf{x}_j$ of length W, this distance is

$$d_{i,j}^{MP} = \sqrt{2W\left(1 - \frac{QT_{i,j} - W\,\mu_i\,\mu_j}{W\,\sigma_i\,\sigma_j}\right)}, \tag{2}$$

where μ_i, μ_j and σ_i, σ_j are the means and standard deviations of the two subsequences, and $QT_{i,j}$ is their dot product. The distance $d_{i,j}^{MP}$ equals the Euclidean distance between the two subsequences after each is standardized to zero mean and unit variance, so it compares shapes, not levels or volatilities. Consider, for example, a quiet utility stock and a volatile technology stock whose 60-day trajectories have the same shape: the two trajectories are close, even though one stock moves much more than the other.

We write the distance between 60-day trajectories $n = (i, t)$ and $n' = (j, s)$ as $d_{n',n}$.[4] A smaller $d_{n',n}$ means that the two trajectories have more similar dynamics. Throughout the paper, "L2 distance" refers to this z-normalized distance $d_{n',n}$.

The Matrix Profile algorithm uses the Fast Fourier Transform (FFT) to compare subsequences within one time series or between two time series. Our comparisons instead run *across* the whole cross-section of stocks (about 1.3 million trajectories) and require more than a trillion distance calculations, including comparisons with the random-walk benchmark series. We therefore compute the distances with GPU-accelerated matrix operations, using the `CuPy` library.[5]

2.3 Random Walk Benchmark

Two trajectories can look alike by chance. How close would a trajectory's neighbors be if returns had no recurring structure? To answer this question, we build a benchmark from shuffled returns, the random-walk (or shuffle) benchmark. Within each calendar year, we pool all daily returns of

[4] Because both vectors are standardized, this distance is a monotonic function of the Pearson correlation. We standardize with the sample standard deviation (divisor 59), so $d_{n',n}^2 = 2 \times 59 \times (1 - \hat{\rho}_{n',n})$; this equals the Matrix Profile distance of Equation (2) up to the constant $\sqrt{59/60}$, which leaves every ranking and the PDR unchanged.

[5] All GPU computations are distributed across three NVIDIA GPUs, running CUDA 12.8 and CuPy 13.6.0. Distance matrices are stored in single-precision floating-point (`float32`) format, retaining approximately 7 significant decimal digits of precision.

all stocks and randomly reassign them across all stock-date observations. Formally, let

$$\mathcal{R}_y = \left\{ r_{j,t} : \text{stock } j \text{ is in the sample on trading day } t \text{ in calendar year } y \right\} \tag{3}$$

denote the set of all daily returns of all stocks in year y, and let $N_y = |\mathcal{R}_y|$ be the number of returns that year. We draw a random permutation σ of $\{1, 2, \ldots, N_y\}$ and assign the $\sigma(k)$-th element of $\mathcal{R}_y$ to the k-th stock-date observation. Each observation (j, t) in year y therefore receives a return drawn uniformly at random, without replacement, from that year's pool.

We repeat the shuffle four times to create four independent random-walk series. The first three series ($m = 1, 2, 3$) form the benchmark for the PDR denominator (Section 2.4), and the fourth series ($m = 4$) stands in for the real data in the placebo test.

The shuffle keeps each year's pooled distribution of daily returns, including its fat tails and its overall spread, so differences between years remain. It removes three things: the time order of each stock's returns, and with it volatility clustering and serial dependence; the differences across stocks, such as their different volatilities; and the alignment of stocks on the same day, and with it market-wide comovement. Every synthetic stock-day is a random draw from that year's pooled returns. This null is deliberately more restrictive than the random-walk hypothesis, which allows heteroskedasticity, jumps, and differences across stocks, so real data can in principle differ from the benchmark for reasons other than recurring shapes. The benchmark is therefore a *normalization device* first; as a null for recurrence, it is a low bar. Section 2.4 sets out what it normalizes. Section 3.3 also tests the persistence against a much stricter null, the stationary block-bootstrap surrogate, which keeps each stock's factor exposures, its own residual distribution, and the short-range dependence of its residuals, such as volatility clustering. The persistence results survive it.

2.4 Pattern Distance Ratio (PDR)

The PDR compares a trajectory's nearest neighbors in real data with its nearest neighbors in the random-walk benchmark. The idea is simple: if a trajectory's shape truly recurs, its neigh-

bors among real trajectories should be systematically closer than its neighbors among random trajectories.

We organize the comparisons in annual blocks. Each trajectory belongs to the calendar year in which its *formation window* ends, and $\mathcal{D}_y$ denotes the set of trajectories whose window ends in year y, for $y = 1971, 1972, \ldots, 2025$. For each block $\mathcal{D}_y$, we compute two types of distance matrices:

- **In-sample:** L2 distances between each trajectory $n \in \mathcal{D}_y$ and every trajectory of the same year or the preceding four years, $n' \in \mathcal{D}_{y-j}$ for $j = 0, 1, \ldots, 4$.
- **Out-of-sample:** L2 distances between each trajectory $n \in \mathcal{D}_y$ and every trajectory of the following five years, $n' \in \mathcal{D}_{y+j}$ for $j = 1, 2, \ldots, 5$.

The persistence analysis uses these comparisons in two ways. The *window* specifications are the 1-Year, 2-Year, 3-Year, and 5-Year columns of Table 2 and of Table 5, Panels A and B. In them, a trajectory formed in year y has its in-sample statistics computed against the trajectories of years $y - h + 1$ through y, and its out-of-sample statistics against the trajectories of years $y + 1$ through $y + h$, for $h = 1, 2, 3, 5$. A longer window therefore raises two things at once: the time distance between the two comparison sets *and* the size of each comparison set. In the *fixed-block, varying-lag* specification (Panel C of Table 5), the comparison set stays at one year, $\mathcal{D}_{y+g}$, and only the lag g changes, which isolates the effect of time distance.

We compute each distance matrix four times: once with real standardized returns for both trajectories (the Real-to-Real distances), and once for each benchmark series $m = 1, 2, 3$, with real query trajectories and random candidate trajectories (the Real-to-Random distances). Trajectories from different stocks can also look alike simply because they move with common systematic factors. To limit this effect, we never compare two trajectories that share the same formation window, so the formation windows of any two compared trajectories start at least one month apart.

For each real trajectory n, we find its $K = 100$ nearest neighbors by the distance in Equation (2) and compute two mean distances:

1. **Real-to-Real Distance** ($\bar{d}_n^{real\to real}$): the mean L2 distance to the K closest other real trajectories,

$$\bar{d}_n^{real\to real} = \frac{1}{K}\sum_{k=1}^{K} d_{n,\,\nu_k^{real}}, \tag{4}$$

where ν_k^{real} is the k-th nearest neighbor of n among the other real trajectories.

2. **Real-to-Random Distance** ($\bar{d}_n^{real\to rand}$): the mean L2 distance to the K closest trajectories of each random series $m \in \{1, 2, 3\}$, averaged across the three series,

$$\bar{d}_n^{real\to rand} = \frac{1}{3}\sum_{m=1}^{3} \bar{d}_n^{real\to rand,m}, \quad \text{where} \quad \bar{d}_n^{real\to rand,m} = \frac{1}{K}\sum_{k=1}^{K} d_{n,\,\nu_k^{rand,m}}, \tag{5}$$

and $\nu_k^{rand,m}$ is the k-th nearest neighbor of n among the trajectories of random series m.

The **Pattern Distance Ratio (PDR)** of trajectory n is

$$PDR_n = \frac{\bar{d}_n^{real\to real}}{\bar{d}_n^{real\to rand}}. \tag{6}$$

If $PDR_n < 1$, the neighbors of trajectory n are closer in real data than in random data, which is consistent with recurrence. If $PDR_n > 1$, the trajectory is less similar to other real trajectories than to random trajectories, which suggests that it is relatively unique in the real data.

Why Normalize by the Random-Walk Benchmark?

Why not simply rank trajectories by the raw distance $\bar{d}_n^{real\to real}$? Because the raw distance changes for three reasons unrelated to genuine recurrence. First, some shapes are common even in random data, so trajectories with these shapes have close neighbors whether or not they recur. A ranking on the raw distance would therefore also sort trajectories by how common their shape is by chance. Second, the number of U.S. stocks, and hence of candidate neighbors in each annual block, changes a great deal over the 55-year sample. Nearest-neighbor distances shrink as the comparison set grows. Third, the distribution of daily returns differs across years, so the shapes that are common by chance also change from year to year.

The denominator controls for all three effects. The random trajectories come from the same pool of returns in the same calendar year, are standardized in the same way, and are searched with the same K nearest neighbors over a comparison set of the same size. Because nearest-neighbor distances are inversely related to local density (Loftsgaarden and Quesenberry, 1965), the PDR has a natural reading as a *local density ratio*: $PDR_n < 1$ means that the region of trajectory space around n is more crowded in real data than chance would imply.

PDR Persistence

Does a trajectory that recurs strongly in one period also recur strongly later? Across the trajectories $n \in \mathcal{D}_y$, we compute the Pearson and Spearman rank correlations between the in-sample PDR (PDR_n^{in}) and the out-of-sample PDR ($PDR_n^{out,h}$). In shuffled data, these correlations would be close to zero; a positive correlation in real data suggests that recurrence is a persistent characteristic that carries over to new data. Because the trajectory is the same in both PDRs and only the trajectories it is compared with move forward in time, what persists is a property of the region of trajectory space around it: how crowded that region is, relative to the benchmark. We report both the pooled correlation across all years and the year-by-year correlations.

PDR Quintile Transition Matrices

The correlations summarize persistence but not where trajectories move. To check whether trajectories also keep their rank, we build transition matrices based on PDR quintiles. For each window length h (one, two, three, or five years), we sort the trajectories of each formation year into five quintiles by their in-sample PDR_n^{in} (Q1 = lowest PDR, or strongest recurrence; Q5 = highest PDR, or weakest recurrence), and separately by their out-of-sample $PDR_n^{out,h}$. The transition matrix $\mathbf{P}^h$ has entries

$$P^h_{q_1,q_2} = \Pr\left(\text{out-of-sample quintile} = q_2 \;\middle|\; \text{in-sample quintile} = q_1\right), \tag{7}$$

estimated as the fraction of trajectories in in-sample quintile q_1 that fall into out-of-sample quintile q_2. Under the null hypothesis of no persistence, every entry equals $1/5 = 0.20$, so diagonal entries above 0.20 mean that trajectories tend to stay in the same quintile over time.

3 Recurrence in Trajectory Space

This section presents the central finding of the paper: how strongly a return trajectory recurs is a persistent characteristic, and this persistence is absent from random data. Using the exhaustive distance comparison of Section 2.2, Section 3.1 shows that the PDR carries over from one period to the next, and Section 3.2 confirms this with transition matrices. Section 3.3 shows that the result survives a stricter benchmark, a strict neighbor rule, controls for volatility, and a shorter trajectory length, and Section 6.1 shows that it holds in twelve international equity markets.

3.1 Pattern Distance Ratio (PDR) Persistence

How much recurrence is there to persist? Table 2 reports the mean PDR of each in-sample quintile. Sorting on PDR produces a spread under any data-generating process, so the informative comparison is with the placebo. With one-year windows, the most recurrent quintile averages 0.982 and the least recurrent 1.038, a spread of 0.056. In the placebo, the two averages are 0.990 and 1.010, a spread of 0.020. The mean over all trajectories is 1.010: on average, a trajectory's real neighbors are slightly farther away than its neighbors in the shuffled series, for the reasons given in Section 2.3. The evidence for recurrence is therefore the spread of the PDR and, below, its persistence, not its level.

Which trajectories, and which neighbors, lie behind these numbers? The five quintiles hold similar stocks: their median market equity, size and book-to-market quintiles, illiquidity, price, and exchanges differ little (Table 3). The most recurrent quintile owes its low PDR to closer real neighbors, not to a more distant benchmark (Table 4, Panel A). A trajectory's 100 nearest neighbors come from about 98 different stocks, and the share from its own industry is the share that random draws would give, both in the same year and in the next (Panel B). The neighbors

found in the next year, moreover, include the stocks of the same-year neighbors only slightly more often than random draws would (4.5 against 4.2 percent). Recurrence is therefore not a matter of the same firms, or of firms in one industry, resembling one another: it is a property of regions of trajectory space.

Does a trajectory that recurs strongly in one year also recur strongly in later years? Figure 2 and Table 5 show that it does. With one-year windows, the pooled correlation between a trajectory's in-sample and out-of-sample PDR is 0.308. The yearly Pearson correlation is positive in all 54 formation years, recession years included, with a mean of 0.317 (Figure 3). Its strength varies over time: it is highest in the 1990s and lowest in 2009.

Could the method itself create this persistence? To check, we repeat the entire PDR computation with the real data replaced by a fourth independent random-walk series (Section 2.3), which by construction has no recurrence beyond chance. In the placebo, both the query trajectories and their neighbors in the numerator come from this series, and the denominator uses the same three benchmark series as before. The placebo's pooled correlation is 0.011, roughly thirty times smaller than in real data, and its out-of-sample decile curve in Figure 2 is flat. The distance metric, the z-normalization, and the comparison design therefore cannot by themselves create persistence of the size we find in real data.[6]

How quickly does persistence fade? Panel C of Table 5 holds the comparison blocks at one year and increases only the distance between them. The correlation falls only from 0.308 one year apart to 0.270 three years apart and 0.257 five years apart, keeping about 83 percent of its one-year value. If recurrence came from short-lived factors, such as temporary liquidity shocks or brief investor attention, it would fade much faster.

Note that longer windows give higher correlations, rising to 0.510 with three-year windows and 0.607 with five-year windows (Panel A). This rise does not mean that persistence grows with distance, because distance alone lowers the correlation (Panel C). A longer window also gives each PDR a larger comparison sample, drawn from more calendar years. The nearest neighbors

[6]The placebo is a single realization, so its correlation is itself a random variable and the informative comparison is one of magnitude. Section 3.3 tests the persistence against a much stricter block-bootstrap null.

then lie closer to the trajectory, so each PDR averages over a smaller region of trajectory space, and they come from more calendar years, so variation specific to one year averages out. Either can raise the correlation.

3.2 Transition Matrices and PDR Stability

Do trajectories also keep their rank? Table 6 shows that they do. For one-year windows (Panel A), all five diagonal probabilities exceed 0.20, the value expected if in-sample rank carried no information, and each exceeds its counterpart in the placebo (Panel E).

Persistence is strongest at the two ends of the distribution. With one-year windows, trajectories with the lowest PDR (Q1, highest recurrence) stay in Q1 with probability 0.267, and trajectories with the highest PDR (Q5, lowest recurrence) stay in Q5 with probability 0.395 (0.226 and 0.225 in the placebo), compared with 0.214–0.226 for the middle quintiles. The diagonal becomes much stronger as the windows get longer: with five-year windows (Panel D), the two probabilities are 0.348 and 0.571.

Movements across quintiles are gradual. In every row of Panels A–D, the probability of moving to another quintile falls steadily as that quintile gets farther from the diagonal in either direction. For example, in Panel A, a Q1 trajectory moves to Q2 with probability 0.232 but jumps to Q5 with probability only 0.117. The placebo has no such ordering: in Panel E a Q1 trajectory reaches Q5 with probability 0.219, almost the same as the 0.226 with which it stays in Q1. What separates the real data from the null is therefore the shape of the whole row, not only the height of its diagonal. The PDR changes slowly rather than abruptly, as a stable characteristic should.

Persistence is not confined to the two extremes. Every diagonal probability in Panels A–D exceeds 0.20, and the middle quintiles retain 0.214 to 0.226 at one-year windows, against 0.209 to 0.213 in the placebo, and 0.239 to 0.266 at five-year windows. Persistence is therefore a property of the organization of trajectory space rather than of a handful of isolated recurring patterns: trajectories keep their place in the ordering, and when they move, a nearby quintile is a likelier destination than a distant one. The return forecast of Section 5 draws its landmark trajectories from the recurrent region only, but the stability documented here runs across the

whole distribution.

3.3 Robustness

Is the persistence an artifact of the random-walk benchmark? The shuffle removes each stock's own time-series structure, the differences across stocks, and their comovement (Section 2.3), so real trajectories could look close to one another merely because of one of these features, such as volatility clustering. We therefore rebuild the benchmark under a much stricter null: a stationary block bootstrap (mean block length of 10 trading days) applied to each stock's residuals from the six-factor model (FF6: the five Fama–French factors plus momentum). This surrogate keeps each stock's estimated factor exposures and the realized factor returns, so comovement through the factors is unchanged. It also keeps each stock's own residual distribution and the short-range dependence of its residuals, such as volatility clustering over a few days. Residuals more than a few weeks apart usually come from different, randomly chosen parts of the stock's history. The persistence survives. The mean year-by-year correlation is 0.247, compared with the 0.317 baseline (Table A2). With a mean block length of 20 trading days, the blocks keep longer stretches of each stock's residuals in their original order. The correlation is then 0.262, slightly higher than with 10-day blocks. Keeping more of the residual dependence in the null does not weaken the persistence.

Is it an artifact of comoving neighbors? A trajectory's 100 nearest neighbors may be dominated by peers from the same period, industry, size group, and value group. We therefore apply a strict neighbor rule: we keep at most one neighbor in each size × book-to-market × 48-industry × month cell before taking the 100 nearest. Under this rule, persistence is stronger on both measures (Pearson 0.350 versus 0.317, Spearman 0.250 versus 0.233). Under both the strict neighbor rule and the block-bootstrap null, persistence remains far above the random-walk placebo, whose mean year-by-year correlation is 0.014 (Table A2, Panel A).

The block-bootstrap null also changes which tail persists more. Under the shuffle benchmark, Q5 retention exceeds Q1 retention at every window length (Table 6, Panels A–D). When we sort trajectories on the surrogate-based PDR, this ordering reverses at the one-year window (Table A2,

Panel B). Q1 retention rises from 0.267 to 0.339, and Q5 retention falls from 0.395 to 0.257. Both tails still persist more than the middle quintiles, whose retention is 0.215–0.220. The fall in Q5 retention suggests that part of the baseline persistence of *uniqueness* comes from features that the surrogate keeps and the shuffle removes (Section 2.3).

Could PDR persistence simply restate the persistence of volatility? Even after standardization, volatile stocks may trace unusual-looking trajectories, and volatility itself is highly persistent. Table A3 shows that this is not the explanation. Adding formation-window idiosyncratic volatility to a Fama–MacBeth regression of out-of-sample on in-sample PDR leaves the coefficient on in-sample PDR almost unchanged (0.304 versus 0.317), and adding size, reversal, and the trajectory's trend leaves it at 0.304. Persistence is also present within every volatility tercile: the mean year-by-year correlation ranges from 0.215 among the least volatile trajectories to 0.389 among the most volatile.

The result also does not depend on the 60-day trajectory length. With 30-day trajectories, the correlation between in-sample and out-of-sample PDR ranges from 0.330 to 0.333 (Table A4). Taken together, the checks above suggest that PDR persistence goes beyond short-range dependence, factor comovement, and the persistence of volatility. Table 7 reports the directional counterpart of this persistence, which Section 4 analyzes.

4 Neighbor Outcomes and Return Predictability

Having shown that trajectory space has persistent local structure (Section 3), we now ask whether the outcomes of similar past trajectories carry information about future returns. This is the direct form of the question: it compares the outcomes of a trajectory's in-sample neighbors with those of its out-of-sample neighbors, with no forecasting model fitted and no portfolio formed. The trajectory's own later return is not used. Section 5 asks the same question through a return forecast; this section asks it of the geometry itself. We measure predictability with the neighbor-outcome score: the mean future Fama–French 25-adjusted (FF25-adjusted) return of each trajectory's nearest neighbors divided by its standard error (defined formally in Online

Appendix B, where we explain why we treat it as a ranking score rather than a test statistic). We compute the score in-sample (T_n^{in}) and out-of-sample ($T_n^{out,h}$) on the same comparison windows as the PDR (Section 2.4), the one-, two-, three-, and five-year windows of Table 5.

A simple test is whether a trajectory's score keeps its sign out of sample. Table 7 reports the sign preservation rate: the share of trajectories whose directional prediction (the sign of T_n) is the same in-sample and out-of-sample. The independence-adjusted null is the rate expected if the in-sample and out-of-sample signs were independent. A rate above it means that the direction of the returns that followed a region of trajectory space tends to repeat in later years, which is the sense in which location carries information about subsequent returns. Panel A sorts all trajectories by the strength of their in-sample score. In every window, sign preservation exceeds the independence-adjusted null in every bin and rises with the score: from 50.40–50.65% for $|T| < 1$ to 52.75–53.81% for $|T| \geq 2.58$, while the null stays between 49.99% and 50.55%. Panel B shows that the information is not confined to the most recurrent trajectories: among high-$|T|$ trajectories, the rate is 51.8–53.7% across the PDR quintiles, above the null of about 50.2–50.3% in every cell.

Two features of the table matter for what follows. The rate rises with the in-sample score in every window, so the score orders trajectories by how reliably their direction survives: the geometry supplies not only a direction but a measure of how much to trust it. And the direction survives throughout the space, not only in its recurrent region: the rate is above the null in all five PDR quintiles. Even so, Q5, where trajectories recur least, has the lowest or second-lowest rate in every window. Recurrence and directional information are therefore related but distinct properties of a location.

The result holds across specifications. Across all four combinations of trajectory length (30 and 60 days) and holding period (20 and 30 days), high-$|T|$ sign preservation is 50.7–53.3% and exceeds the independence-adjusted null in every PDR quintile (Table A5). Under the strict neighbor rule of Table A2, high-$|T|$ sign preservation is 52.3–53.8%, above the null in every quintile and above the baseline rate in four of the five (Table A6). In both tables, Panel A sorts all trajectories by $|T|$ as Table 7 does: the rate exceeds the null in every bin, and trajectories with

low scores ($|T| < 1.96$) preserve sign at 50.4–51.8%, above the null but much closer to it. Taken together, the high-score rate exceeds the independence-adjusted null in every high-score cell of Tables 7, A5, and A6. The baseline rests on 134,545 high-score trajectories and the strict rule on 105,857.

Together with the persistence evidence of Section 3, these results indicate that the geometry is not merely persistent: location within it is informative about subsequent returns, and more so when the score is stronger. Section 5 turns this trajectory-level evidence into a tradable long–short strategy.

5 A Trajectory-Space Portfolio Strategy

This section asks how much predictive information can be extracted when we use the geometry of trajectory space more fully, by building a tradable long–short strategy. The portfolio is a demanding test of the geometry rather than a proposed trading rule: it asks whether the information in trajectory space is systematic enough to survive aggregation into return forecasts. The idea is simple: a stock's expected return is estimated from the realized outcomes of past trajectories that look like its current one. Section 4 looked only at the outcomes of a trajectory's 100 nearest neighbors. Here we extend that local view: the forecast uses the trajectory's similarity to many locations in trajectory space at once, and a ridge regression decides how much each location counts. Section 5.1 describes the estimator, and Section 5.2 reports its risk-adjusted performance. Section 5.3 measures the strategy's turnover and break-even trading costs, and Section 5.4 asks what the predictability means. Section 6.2 then applies the estimator to twelve international equity markets as an out-of-sample test.

5.1 Methodology

The estimator, which we call the engine, works in the trajectory space of Stage I and uses its distance metric. It is an approximate kernel ridge regression in this space, so each forecast is a similarity-weighted combination of past outcomes. The construction has four steps.

1. **Landmarks.** We fix a set of *landmark* trajectories, the *landmark set* of Online Appendix B. From the trajectories of the previous five years, we keep those with a strong neighbor-outcome score (Section 4) and a low PDR (Section 3), both measured against the three years before each trajectory's own year. The score must be at least the conventional cutoff in absolute value ($|T| \geq 1.96$), and the PDR must be at or below the 20th percentile of this pool's PDR distribution. The grid therefore sits in the most recurrent part of trajectory space. Every qualifying trajectory is a landmark, with no cap on their number; Table A7 reports the counts.

2. **Features.** We represent each 60-day standardized return trajectory by its location in trajectory space: the vector of its Gaussian-kernel similarities to the landmarks, a landmark basis of the kind used in low-rank kernel approximations (Williams and Seeger, 2001).

3. **Prediction.** We predict the FF25-adjusted forward return (Online Appendix B) by ridge regression on this similarity vector, estimated once per test year on an expanding, purged window. The training sample holds every trajectory whose formation window ends before December of the previous year, which leaves a one-month embargo ahead of the test year. Before the regression, we residualize the return each month on the stock's five-day and 21-day reversal characteristics: the compound returns over the last five and the last 21 trading days of its formation window, each winsorized at the 1st and 99th percentiles within the month. The ridge target is the within-month percentile rank of that residual, mapped through the inverse of the normal distribution function (a rank-Gaussian target).

4. **Signal.** In the spirit of the residual-return construction of Da et al. (2014), we orthogonalize the prediction every month to the same two characteristics, and its within-month rank is the trading signal we use throughout.

We keep the full high-dimensional similarity vector and regularize it with ridge, rather than selecting a few features, following the virtue-of-complexity logic for return prediction (Kelly et al., 2024).

What role do the landmarks play? They fix the basis in which the ridge regression is expressed. Directional information is present before any basis is chosen: the sign-preservation evidence of Section 4 uses no landmarks (Table 7).

Could the forecast be momentum in disguise? Nagel (2025) warns that the virtue-of-complexity logic can be misleading. His main setting is timing the aggregate stock market with persistent predictors and far more kernel features than training months. He shows that his argument extends in part to a cross-sectional panel, where the kernel smoothing runs across stocks as well as across time. In a short training window, similarity in the predictor space mainly reflects closeness in time. The kernel forecast then reduces to a recency-weighted average of recent returns, essentially a momentum strategy. Three features of our design work against this. First, stocks are ranked within each month, so a recency component common to all stocks cancels out of the long–short portfolio. Second, the training sample is large relative to the number of features. The window starts in 1971, expands every year, and weights old and recent trajectories equally. Predictions begin only once it holds 50,000 trajectories, while the grid never exceeds 4,780 landmarks. Third, and closest to the cross-sectional channel, a stock's recent trajectories need not resemble its current one. The distance compares two trajectories day by day, so trajectories of the same stock taken a month apart line up different trading days at every position. Section 5.2 tests the signal directly against momentum and moving-average rules.

At the end of each month, we sort stocks into quintiles by the signal and trade a long–short portfolio that buys the top quintile (Q5) and sells the bottom quintile (Q1), with both equal and value weights, over the out-of-sample period 1975–2025.

5.2 Performance

Excess returns and FF6 alphas both rise monotonically from Q1 to Q5, for equal- and value-weighted portfolios alike (Table 8). Equal-weighted, the excess return rises from 3.77% per year in Q1 to 12.53% in Q5.

The strategy earns a large and highly significant alpha under every factor model, for both equal- and value-weighted portfolios (Table 9). The FF6 alphas are 8.01% and 6.16% per year,

respectively.

Is the alpha just short-horizon reversal? The question is natural, because the engine reads a 60-day return trajectory whose final days are the most recent week (Jegadeesh, 1990; Lehmann, 1990). The alpha survives adding French's published *monthly* short-term reversal factor, ST-Rev (8.21% equal-weighted and 6.18% value-weighted). It also survives when we add, on top of ST-Rev, a *weekly* reversal factor (WST-Rev) that we build inside our universe with the strategy's portfolio rule: the alpha is then 6.63% equal-weighted and 5.98% value-weighted.

Is the evidence truly out-of-sample? In the statistical sense it is: no forecast uses information realized after its formation date. The training windows expand, are purged, and have an embargo, and the landmarks are chosen point-in-time. A stronger concern is data snooping: an estimator with many design choices can fit a sample by chance. Two tuning parameters follow standard rules. The kernel bandwidth is set by the median heuristic and the ridge penalty by generalized cross-validation. Both rules use only data from before the test year, so neither value is chosen by out-of-sample performance. The international evidence is out-of-sample by construction. Section 6.2 applies the same specification to twelve international markets, apart from two details that the international data require. There, the global country-average portfolio earns a significant raw return over 2001–2025.

Does the signal simply repackage known predictors? Because the engine reads return trajectories, it may pick up technical rules and characteristics already known to predict returns. This is the comparison that the conventional approach of the Introduction suggests: several of the regressors summarize the same 60-day path, namely the moving-average rules, the 5- and 21-day reversal, the largest daily return, and idiosyncratic volatility. Table 10 tests this directly, with every regressor winsorized and standardized within the month so that slopes are comparable. Moving-average rules built from the same 60-day window do predict returns in our sample, and so do reversal, momentum, idiosyncratic volatility, size, and liquidity. Yet with all of them included, the signal's equal-weighted Fama–MacBeth coefficient is 0.138, with a t-statistic of 8.4, and adding the 5-day reversal characteristic does not reduce it (0.169, $t = 9.2$). In value-weighted regressions (Table A8), the coefficient is 0.074 ($t = 3.0$), smaller but still significant at the 1

percent level.

5.3 Implementability and the Cost of Arbitrage

The alphas above are gross of trading costs, and the strategy trades actively: under monthly rebalancing, it replaces 82 to 84 percent of each leg every month. How large a trading cost can the strategy bear? Following the literature on anomaly trading costs (Novy-Marx and Velikov, 2016; Chen and Velikov, 2023), we compute turnover after letting the previous month's weights drift with their realized holding-period returns, so that passive price changes are not counted as trading.

Table 11 reports, in the form of Grundy and Martin (2001), what the strategy earns, how much it trades, and the trading cost that would exhaust its earnings. We report this break-even cost for two reasons. First, it uses only the alpha and the turnover, so it does not depend on how well spreads can be measured. Second, what decides implementability is the reader's own execution cost (Bajgrowicz and Scaillet, 2012). The break-even cost is a spread per dollar traded, not the cost of borrowing the short leg. The alphas of 8.0% and 6.2% per year are exhausted by a one-way cost of 20.3 basis points equal-weighted and 15.2 value-weighted. Among stocks in the top NYSE-breakpoint size tercile, the hurdle is lower, 13–20 basis points across the three sample periods, because the gross alpha falls while the turnover barely changes. The stricter of Grundy and Martin's two conventions is the cost at which the profit can no longer be distinguished from zero. Under it, the hurdle is 7–15 basis points, and among the largest firms since 2009 it falls to 5.2 basis points value-weighted and 2.5 equal-weighted.

Would realistic costs exhaust the alpha? For most investors they would, and this is an important economic limit on the strategy. Novy-Marx and Velikov (2016) find that few anomaly strategies with one-sided monthly turnover above 50 percent survive their trading costs, and this strategy's turnover is well above that level. The exception is the modern large-cap universe, where realized institutional costs are lower than academic spread estimates (Frazzini et al., 2018). A trader whose costs are in the single digits of basis points would clear the 13–16 basis-point hurdle of the 2009–2025 large-cap portfolio (the $\alpha = 0$ column of Table 11). In that universe,

implementability turns on execution quality. The predictability documented in this paper should be read as evidence about how prices form, not as a claim of implementable arbitrage profits.

5.4 Interpretation

What does the predictability mean? It helps to separate two questions. The first is where it comes from: it may compensate investors for a risk that our factor models miss, or it may reflect mispricing. The second is why it survives: if the strategy were a free lunch, arbitrageurs should trade it away. Our evidence speaks to both, and most directly to the second.

Consider first the source. The two readings make different predictions. A mispricing should be stronger in stocks that are hard to arbitrage, and it should weaken as arbitrage capital grows (Shleifer and Vishny, 1997; McLean and Pontiff, 2016; Stambaugh et al., 2015). A risk premium need not do either. We therefore compare the signal across firm-size terciles (Table 12). Because a raw long–short alpha is automatically larger where returns are more dispersed, we use two scale-free measures of signal quality: the rank information coefficient (rank-IC) and the Sharpe ratio.

The evidence points both ways. Signal quality is higher in small stocks: over the full sample, the rank-IC is 0.026 in the small-cap tercile and 0.020 in the large-cap tercile, and the equal-weighted Sharpe ratio is 1.40 versus 0.71. Among small stocks, this Sharpe ratio also declines over time, from 1.40 over the full sample to 1.07 over 2000–2025 and 0.82 over 2009–2025. Both facts fit a mispricing story. On the other hand, the value-weighted FF6 alpha is no smaller over 2000–2025 and 2009–2025 than over the full sample: it rises from 6.16% to 8.25% and 6.96% (Table 9). Over 2009–2025, the large-cap tercile also has the higher rank-IC (0.022 versus 0.018). Moreover, because of the joint-hypothesis problem (Fama, 1970), an alpha always tests market efficiency and the factor model together, so a risk that our models miss cannot be ruled out. We therefore take no stand on the source.

Consider next why the predictability survives. Here the evidence says more. The strategy trades heavily, and a one-way cost of about 15 to 20 basis points would exhaust its alpha (Table 11). When trading is costly, prices need to reflect information only up to the point where the gain from

trading on it no longer exceeds the cost (Grossman and Stiglitz, 1980; Fama, 1991). Predictability of this kind need not disappear. This view also fits the facts above: over the full sample, the predictability is stronger in small stocks, where trading is more expensive (Novy-Marx and Velikov, 2016). The cost channel binds least for large stocks since 2009, yet the alpha remains significant there (Table 11, Panel B).

In short, whatever its source, the predictability is consistent with a regularity that survives because it is costly to trade. The source could be studied more directly, for example by relating the predictability to investor sentiment (Stambaugh et al., 2012).

6 International Evidence

This section takes the recurrence result and the engine outside the United States. Does recurrence persist in other markets, and does the engine predict returns there? We take the two questions in turn.

6.1 Recurrence Abroad

The persistence is not limited to U.S. equities. We repeat the PDR construction in twelve international equity markets, using daily Compustat Global prices cleaned largely as in Chaieb et al. (2021).[7] The candidate pool is every Compustat Global market that could reach 5,000 trajectories in some year, and the depth rule explained below selects the twelve. Table A11 in Online Appendix E lists them with their exchanges, currencies, and regional factors. The

[7]The data are daily prices from Compustat Global (`g_secd`): common shares (issue type code 0) of the firms headquartered in the market, on the exchange that lists the most of them (the two that list the most in China, France, Germany, India, Japan, South Korea, and Taiwan; Table A11 names those of the twelve markets); a security-description screen that removes preferred shares, warrants and rights, bonds, funds, trusts, REITs, depositary receipts, certificates, and exchange-traded products; the most-traded issue of each firm; and prices from days with an actual trade. Cross-listed securities and exchange-traded products carry quotes on local holidays. A date therefore counts as a trading day only when the number of priced securities is at least 20 percent of the securities priced that month and at least $0.8\tilde{m} - 2\sqrt{\tilde{m}}$, where $\tilde{m}$ is the median count over the 21 nearest dates. Returns compound across the dates dropped. Daily total returns come from adjusted closing prices and total-return factors, and are set to missing on the day a security's quotation currency changes or when they exceed 100 percent in absolute value. Following Shumway (1997), a return of −30 percent is assigned on the first trading day after the last observation of a security delisted for bankruptcy or liquidation. June market equity, converted to U.S. dollars at that day's exchange rate, must be at least US$10 million for a stock to enter the following July–June year.

in-sample/out-of-sample correlation is positive in every market, and in at least 70 percent of the market-years within each market (Table 13). It varies from 0.51 in Hong Kong to 0.09 in China, is significant at the 1 percent level in every market, and exceeds the U.S. mean of 0.317 in seven of the twelve. Section 6.2 applies the full prediction engine to these markets as an out-of-sample test.

Why only twelve markets? The PDR is a *local density* ratio: the distance from a trajectory to its nearest neighbors *within the same market and year*, divided by the same quantity against a benchmark from that market and year. How local this measure is therefore depends on how densely the same-year cross-section fills the neighborhood of each trajectory, not on the length of any stock's history. In a thin market, the nearest neighbors are far away, so each PDR averages over a wide region of trajectory space and blurs the local structure it is meant to measure. This blurring can push the measured correlation toward zero even if structure is present. We therefore apply a depth rule. Each market-year must have at least 5,000 trajectories observed in both the in-sample and the out-of-sample block, and we keep markets with at least five qualifying market-years, so that each market's t-statistic in Table 13 rests on at least five annual correlations. The footnote accounts for every market the rule excludes.[8]

6.2 The Engine Abroad

The most demanding test of a return predictor built on more than fifty years of U.S. data is to apply it to other markets, with its estimation rules frozen at their U.S. values. We apply the engine to the twelve international markets of Table 13, the markets that meet the depth rule of Section 6.1. We follow the design of the literature that transplants anomalies abroad (McLean et al., 2009; Watanabe et al., 2013; Jacobs and Müller, 2020): portfolios formed within each market, regional factor adjustment (Fama and French, 2012), and pooled cross-country

[8]Twenty-five Compustat Global markets have enough listed securities to reach 5,000 trajectories in some year (at least 417 issuers with a price in that year). Fourteen of them have at least one qualifying year; Indonesia (three years) and Turkey (two) fall short of five, and the other eleven, including Germany, have none. Studies that estimate factor exposures stock by stock screen on the length of each stock's own history: Chaieb et al. (2021) keep a stock only if it has at least five years of monthly returns. Stocks in a thin market can meet such a requirement even when the market's same-year cross-section is too sparse for the PDR.

regressions on within-country ranks.[9]

Apart from two details, the estimation rules are the U.S. ones (the landmark rule, the bandwidth and penalty rules, the training window, the target, and the portfolio rules), while each market's landmark set, bandwidth, penalty, and regression coefficients are re-estimated from its own data. The two details differ because the international data require them. The forward return is measured in excess of the equal-weighted average of the market's other stocks, in both the target and the neighbor-outcome score, since no FF25 benchmark exists abroad; and predictions begin once a market's training sample reaches 20,000 trajectories rather than 50,000, because several markets have far fewer trajectories than the United States.[10] Predictions begin between 1993 and 2009 and run through December 2025. Each market's long–short portfolio is evaluated against its regional five-factor model (Fama and French, 2017) augmented with the regional momentum factor of Fama and French (2012).[11]

Tables 14 and 15 give three layers of evidence. First, the equal-weighted long–short alpha is positive in all twelve markets and significant at the 5 percent level in eight; the four exceptions (the United Kingdom, Thailand, Taiwan, and China) have t-statistics below 1.9. Second, the country-average portfolio of Chui et al. (2010), which weights each country's long–short portfolio equally, earns 6.79% per year with equal-weighted stocks ($t = 11.8$) and 4.66% with value-weighted stocks ($t = 5.1$) over 2001–2025.[12] Against French's Developed and Emerging factors, respectively,

[9]Table 13 reports each market's trajectories, stocks per month, and out-of-sample window. The Compustat Global screens of Section 6.1 play the role that the Ince and Porter (2006) and Hou et al. (2011) filters play in Datastream-based studies. The data include delisted securities, which limits survivorship bias. Returns are in local currency for at least 99 percent of daily observations, and because both legs of each long–short portfolio are in the same currency, exchange-rate movements affect the long–short return only at second order.

[10]The reversal residualization and the rank transform of the target are unchanged. Sweden, Malaysia, Thailand, and Australia have 93 to 128 thousand trajectories against 1.3 million in the United States. Code and a replication package are available from the authors.

[11]Following Fama and French (2012), who show that global models fail to price regional portfolios, we use regional rather than global factors from Kenneth French's data library: the Europe, Japan, and Asia Pacific ex Japan series for the developed markets, and the Emerging Markets series for China, India, South Korea, Malaysia, Taiwan, and Thailand. The emerging-market series is built by the same procedure but lies outside the developed-market samples of Fama and French (2012) and Fama and French (2017). No international short-term reversal factor is available; Table 15 instead controls for the prior-month and last-week returns at the characteristic level.

[12]We require at least five active markets in a month, while Chui et al. (2010) require two; Asness et al. (2013) instead weight each market by the inverse return volatility of its benchmark index. Following Chui et al. (2010) and Jacobs and Müller (2020), we treat these raw returns as the main statistic, because French's Developed factors do not

the six Developed markets earn 8.48% (t = 9.8) equal-weighted and 6.01% (t = 3.7) value-weighted, and the six Emerging markets earn 6.36% (t = 7.4) and 4.37% (t = 3.0). Third, in pooled Fama–MacBeth regressions on within-country-month ranks (Table 15), the signal's slope is 0.59% per 20-day holding period (t = 9.8) on its own and 0.54% (t = 9.6) with controls for size, book-to-market, prior-month reversal, and 12–2 momentum. These controls extend the size, book-to-market, and momentum controls of McLean et al. (2009) and Watanabe et al. (2013). Adding the last-week return barely moves it (0.55%, t = 9.7), and so does adding Amihud (2002) illiquidity (0.55%, t = 9.8). Book-to-market is itself priced (0.66%, t = 2.9), so the value effect is present.

Three caveats apply. The $10 million size floor, our own addition beyond the Chaieb et al. (2021) screens, is weaker than the U.S. size and price screens, so the international samples can reach smaller firms than the U.S. sample. The out-of-sample windows also vary in length, from 204 months for Sweden to 396 for Japan. All returns are gross of trading costs. Across markets, the country-average portfolios remain significant on both weightings (Table 14).

7 Conclusion

In this paper, we take a geometric perspective on return dynamics. Rather than compressing return trajectories into predefined signals or patterns, we organize approximately 1.3 million 60-day return trajectories directly through pairwise similarity, constructing a geometric trajectory space in which distance, neighborhood, recurrence, persistence, and location become natural empirical objects. Inspired by the Matrix Profile principle of exhaustive subsequence comparison, our approach searches this space exhaustively and therefore remains assumption-light about the forms that relevant patterns may take. We use this framework to ask two broad questions: whether trajectory space exhibits persistent structure, and whether location within that space contains information about subsequent returns.

To study the first question, we introduce the Pattern Distance Ratio (PDR), which measures

span the six markets in our set that belong to his Emerging universe.

a trajectory's local recurrence relative to a matched random-walk benchmark. We find that recurrence is a persistent characteristic of trajectory space. Trajectories that recur strongly in one period tend to recur strongly in later periods, and this persistence is absent from random-walk data and survives a stricter block-bootstrap benchmark. Persistence is strongest at both ends of the distribution: highly recurrent and highly unique trajectories are the most likely to remain in their respective regions of the space.

The geometric structure also contains information about subsequent returns. The outcomes of similar past trajectories preserve their direction out of sample more often than chance, with stronger scores exhibiting greater persistence. Using the full similarity representation to predict returns produces significant gross abnormal returns unexplained by standard factor models, and the evidence reappears across twelve international equity markets. Transaction costs, however, mostly eliminate the strategy's profitability, placing an important economic limit on implementability.

More broadly, the evidence suggests that return trajectories exhibit persistent organization that is difficult to reconcile with a random-walk benchmark and that location within trajectory space is systematically related to subsequent outcomes. Prediction is one implication of this structure rather than what defines it. Treating return trajectories as points in a geometric space provides a way to study return dynamics through their distances, neighborhoods, recurrence, persistence, and location without requiring the relevant patterns to be specified in advance.

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Figure 1: Overview of the Methodology

The procedure has two stages. *Stage I (Recurrence Identification)* constructs return trajectories, computes the Pattern Distance Ratio (PDR) relative to a random-walk benchmark, and tests whether recurrence persists over time. *Stage II (Return Predictability)* first fixes a basis of landmark trajectories from the previous five years (Section 5.1). It then represents each trajectory by its kernel similarities to the landmarks, predicts returns from these similarities by ridge regression, orthogonalizes the predictions to short-horizon reversal, and forms portfolios from the resulting signal. The dashed box marks the random-walk benchmark (Section 2.3). Section references appear in parentheses.

Alt text: Flowchart of the two-stage procedure: four boxes in the first stage and five in the second, joined by downward arrows. A dashed random-walk box at the side feeds the Pattern Distance Ratio calculation.

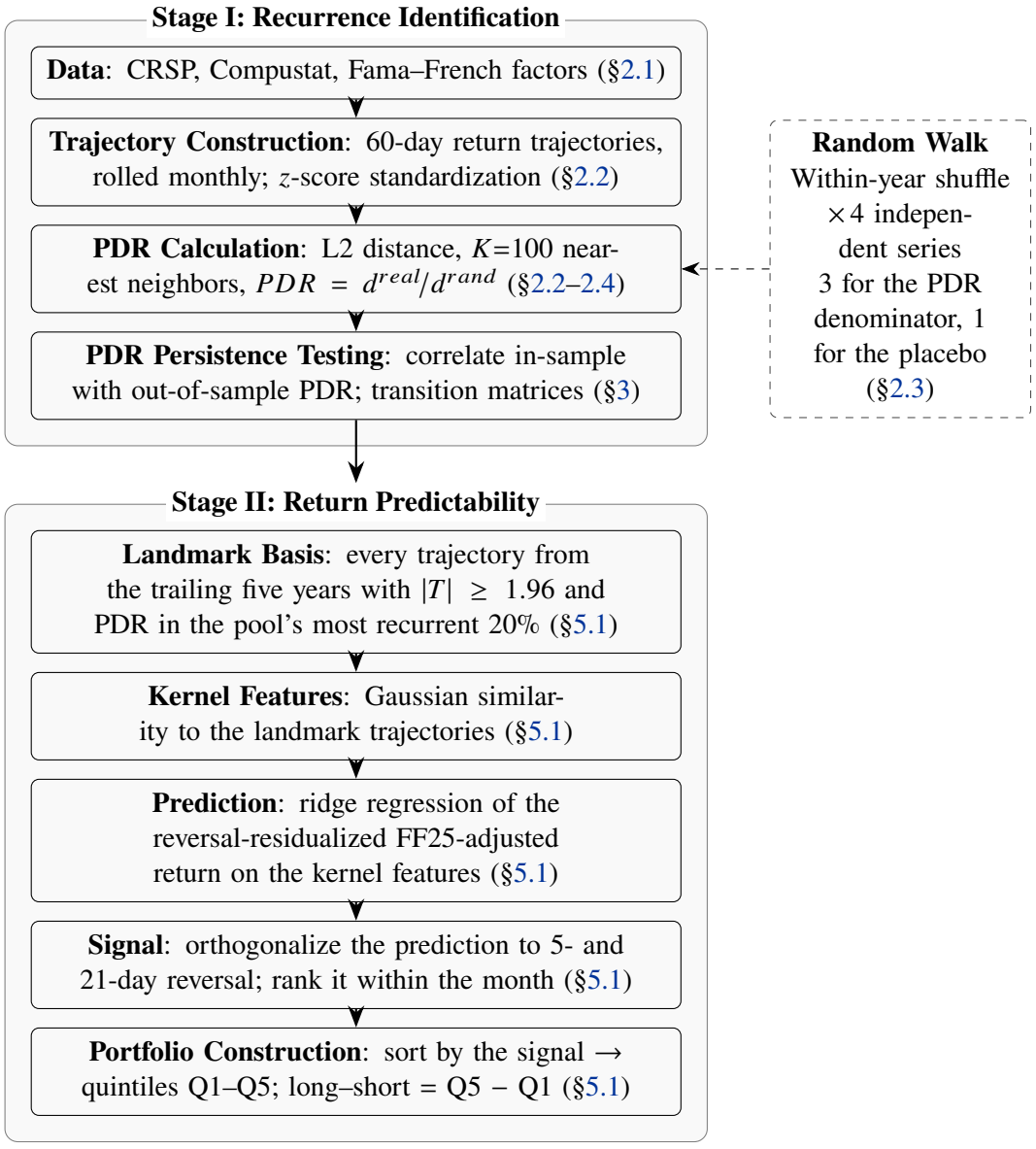


Figure 2: PDR Persistence across Comparison Windows

This figure shows the persistence of the Pattern Distance Ratio (PDR) across comparison-window lengths. Each column is one specification: the 1-Year, 2-Year, 3-Year, and 5-Year windows of Table 5 and a random-walk placebo. The top row plots out-of-sample PDR against in-sample PDR for a random subsample of 3,000 trajectories, with a dashed 45-degree line, dotted gray lines at $PDR = 1$, and the pooled Pearson correlation r. The bottom row sorts trajectories into deciles of in-sample PDR within each formation year and plots the mean in-sample PDR (circles) and the mean out-of-sample PDR (squares) in each decile, with the shaded band showing the gap between them. The in-sample curve rises by construction, since deciles are sorted on it; the out-of-sample curve rises more gently, as expected when the two measures are imperfectly correlated, and steepens as the comparison window lengthens. Under the placebo it is flat. All analyses use data from 1971–2025.

Alt text: Two rows of five panels. For real data, the scatter clouds tilt along the 45-degree line, more tightly for longer windows, and the out-of-sample decile curves rise. For the placebo, the cloud is round and the curve is flat.

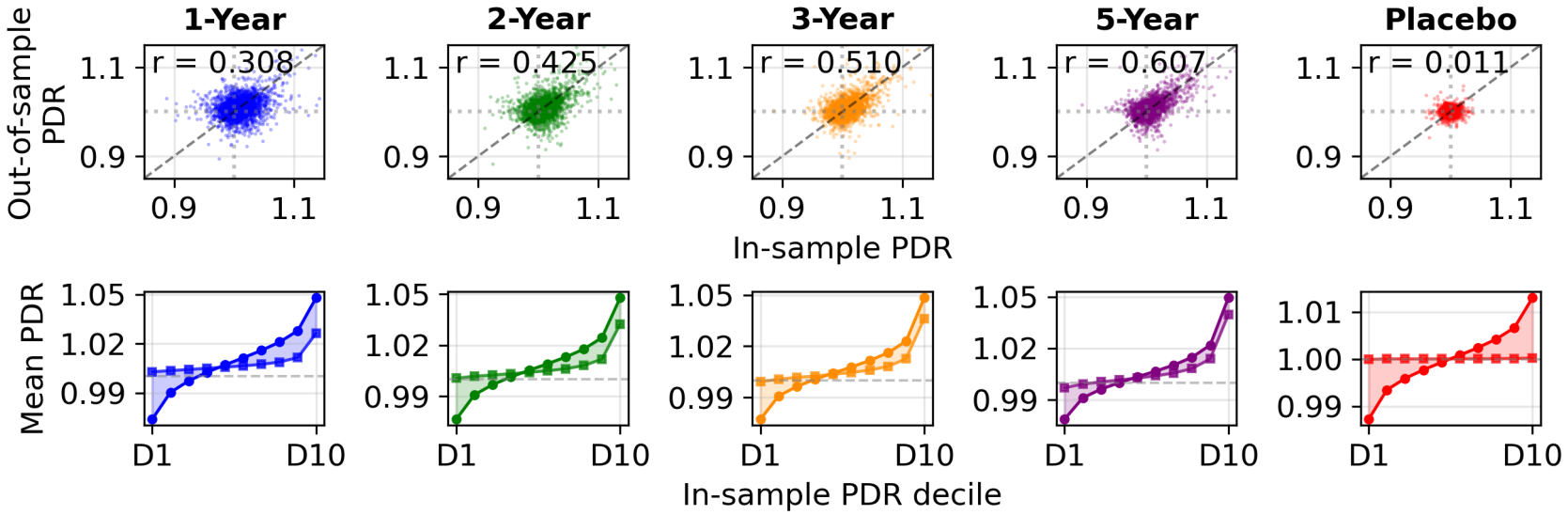


Figure 3: PDR Persistence by Formation Year

This figure plots the Pearson correlation between in-sample and out-of-sample Pattern Distance Ratio (PDR) for each formation year from 1971 to 2024. It uses the 1-Year windows of Table 5. Gray shaded areas mark NBER recession periods. The horizontal dotted line marks the mean of the yearly correlations, 0.317.

Alt text: Line chart, positive in every year. It runs between about 0.16 and 0.35 until 1986, stays above 0.5 from 1991 to 2000 with a peak of 0.65 in 1994, falls to 0.05 in 2009, and then runs between 0.10 and 0.35.

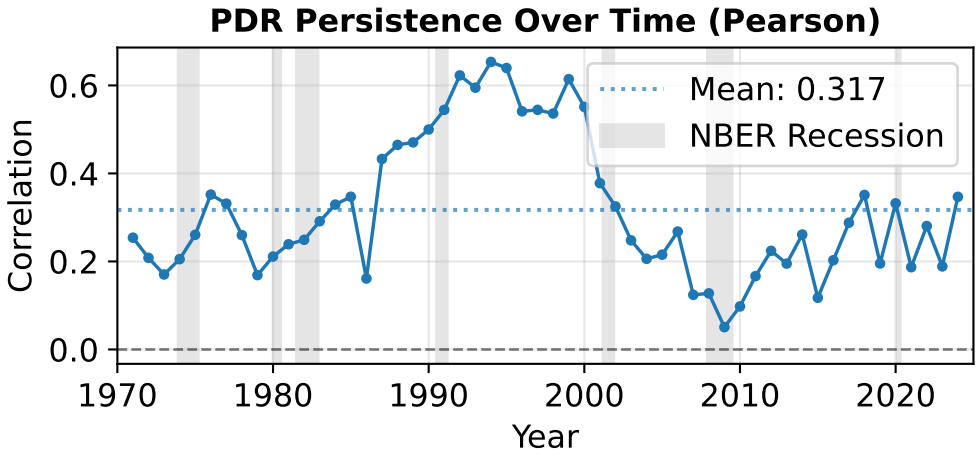


Table 1: Summary Statistics

This table describes the sample of Section 2.1. A trajectory is a stock's 60 daily returns ending on the last trading day of a month, its formation date, and a stock has at most one trajectory a month. Panel A applies the screens of Section 2.1 in turn, over July 1971 to December 2025, and reports the stocks, stock-years (July to June), and stock-months that remain. A firm with several share classes keeps the class with the largest market equity, which carries the firm's total market equity, and the book-equity screen also requires a CRSP–Compustat link and market equity at the end of June and of the previous December. The last row counts trajectories, which require returns on all 60 trading days of the formation window and on the 20 trading days that follow. Panel B reports, by decade of the formation date, the number of trajectories, the average number of stocks per month, the median market equity on the formation date, and the shares of trajectories from NYSE, AMEX, and Nasdaq stocks.

Panel A: Sample construction

	Stocks	Stock-years	Stock-months
CRSP common stocks on NYSE, AMEX, and Nasdaq	25,560	295,997	3,271,048
One share class per firm	25,450	293,393	3,238,277
Positive book equity, two years of Compustat data	21,210	226,740	2,607,537
June market equity ≥ NYSE 10th percentile	12,260	122,238	1,420,416
June price ≥ \$5	11,996	119,009	1,383,283
Trajectories: complete 60-day window and 20-day holding period	11,781	117,422	1,316,273

Panel B: The sample by decade

	1971–79	1980–89	1990–99	2000–09	2010–19	2020–25	All
Trajectories	158,602	222,375	295,840	271,073	229,040	139,343	1,316,273
Stocks per month	1,586	1,853	2,465	2,259	1,909	1,963	2,022
Median ME (\$ millions)	116	250	446	1,013	2,038	3,075	704
NYSE (%)	70.0	60.4	51.6	52.3	53.4	49.1	55.5
AMEX (%)	9.7	8.2	4.9	1.6	0.7	0.4	4.2
Nasdaq (%)	20.2	31.4	43.5	46.1	45.9	50.5	40.3

Table 2: Mean PDR by In-Sample Quintile

This table reports the mean in-sample Pattern Distance Ratio (PDR) of each in-sample PDR quintile (Q1 = most recurrent, Q5 = least recurrent); quintiles are formed within each formation year. In the 1-Year, 2-Year, 3-Year, and 5-Year specifications, a trajectory formed in year y has its in-sample PDR measured against the trajectories of years $y - h + 1$ through y and its out-of-sample PDR measured against the trajectories of years $y + 1$ through $y + h$, for $h = 1, 2, 3, 5$. Both windows must lie within 1971–2025, which leaves 54, 52, 50, and 46 formation years. The *All trajectories* row is the mean over all trajectories. The Placebo column repeats the 1-Year specification on the placebo series, a single realization of the shuffled benchmark. All analyses use data from 1971–2025.

Quintile	1-Year Windows	2-Year Windows	3-Year Windows	5-Year Windows	Placebo
Q1	0.982	0.983	0.984	0.985	0.990
Q2	1.000	0.999	0.999	0.998	0.997
Q3	1.009	1.007	1.006	1.005	1.000
Q4	1.019	1.015	1.014	1.013	1.003
Q5	1.038	1.036	1.035	1.035	1.010
All trajectories	1.010	1.008	1.007	1.007	1.000

Table 3: Characteristics of Trajectories by PDR Quintile

This table describes the trajectories of each in-sample PDR quintile of Table 2 under 1-Year windows (Q1 = most recurrent, Q5 = least recurrent). Quintiles are formed within each formation year, and the statistics pool all 54 formation years. Panel A describes the stock on the trajectory's last day: its market equity, the NYSE size and book-to-market quintiles of its Fama–French 25 portfolio (1 = smallest, lowest), its Amihud (2002) illiquidity as defined in Table A10, its price, and its exchange. Panel B describes the 60 daily returns that form the trajectory: their compound return and standard deviation. Market equity, illiquidity, price, the compound return, and the standard deviation are medians; the quintiles and the exchange shares are means.

	Q1	Q2	Q3	Q4	Q5	All
Panel A: The Stock						
Market equity ($ millions)	691	708	713	698	623	686
Size quintile (NYSE)	2.78	2.81	2.81	2.80	2.70	2.78
Book-to-market quintile (NYSE)	2.65	2.65	2.64	2.64	2.59	2.63
Illiquidity (Amihud)	0.0075	0.0074	0.0071	0.0073	0.0075	0.0073
Price ($)	25.8	25.6	25.5	25.1	24.2	25.2
NYSE (%)	54.9	56.3	56.6	56.4	53.7	55.6
AMEX (%)	3.9	4.0	4.1	4.3	4.7	4.2
Nasdaq (%)	41.2	39.7	39.3	39.3	41.6	40.2
Panel B: The 60 Daily Returns						
Compound return (%)	2.59	2.47	2.28	2.07	1.90	2.28
Standard deviation (%)	2.06	2.06	2.08	2.12	2.35	2.12
Trajectories	259,140	259,106	259,108	259,106	259,130	1,295,590

Table 4: Neighborhoods by PDR Quintile

This table describes the nearest neighbors behind the PDR, by in-sample PDR quintile of Table 2 under 1-Year windows. Panel A reports the two parts of the in-sample PDR: the mean distance from a trajectory to its 100 nearest real trajectories (the numerator) and to its 100 nearest trajectories of the shuffled benchmark, averaged over the three series (the denominator). Panel B describes the 100 nearest real trajectories in the same year (in-sample) and in the next year (out-of-sample): the number of distinct stocks among them, the share from the trajectory's own stock, the share from its industry (the 48 industries of the strict neighbor rule of Table A2) and that share if the 100 were drawn at random from the candidate trajectories. The last two rows report the share of the out-of-sample neighbors' stocks that are also among the in-sample neighbors' stocks, and that share if the out-of-sample neighbors were drawn at random. All statistics are means over trajectories, pooled over the 54 formation years.

	Q1	Q2	Q3	Q4	Q5	All
Panel A: Mean Distance to the 100 Nearest Trajectories						
Real trajectories (numerator)	8.364	8.517	8.582	8.629	8.585	8.535
Shuffled benchmark (denominator)	8.516	8.518	8.503	8.473	8.275	8.457
Panel B: The 100 Nearest Real Trajectories						
In-sample neighbors (same year)						
Distinct stocks	98.4	98.2	98.1	98.1	98.0	98.2
Same stock (%)	0.05	0.05	0.05	0.05	0.05	0.05
Same industry (%)	4.7	4.5	4.5	4.4	4.5	4.5
Same industry, at random (%)	4.5	4.5	4.5	4.4	4.4	4.5
Out-of-sample neighbors (next year)						
Distinct stocks	98.1	98.2	98.2	98.2	98.1	98.2
Same stock (%)	0.05	0.05	0.05	0.05	0.05	0.05
Same industry (%)	4.6	4.6	4.5	4.5	4.5	4.6
Same industry, at random (%)	4.5	4.5	4.5	4.5	4.4	4.5
Stocks shared with in-sample (%)	4.5	4.5	4.5	4.5	4.6	4.5
Stocks shared, at random (%)	4.3	4.2	4.2	4.2	4.1	4.2

Table 5: PDR Persistence Comparison

This table compares the persistence of the Pattern Distance Ratio (PDR) with that in a random-walk placebo test. The 1-Year, 2-Year, 3-Year, and 5-Year specifications and the Placebo column are those of Table 2. Panel A reports the pooled correlations between in-sample and out-of-sample PDR. Panel B reports the mean of the Spearman correlations computed separately for each formation year. Panel C isolates the effect of temporal distance alone. It holds the comparison block fixed at one year and varies only the lag g. That is, it correlates the in-sample PDR of annual block $\mathcal{D}_y$ with the out-of-sample PDR measured against $\mathcal{D}_{y+g}$ for $g = 1, 2, 3, 5$. All analyses use data from 1971–2025.

Panel A: Pooled Correlations					
Metric	1-Year Windows	2-Year Windows	3-Year Windows	5-Year Windows	Placebo
Pooled Pearson r	0.308	0.425	0.510	0.607	0.011
Pooled Spearman r	0.251	0.317	0.375	0.454	0.008
N (trajectories)	1,295,590	1,267,818	1,229,810	1,146,275	1,295,608
Panel B: Year-by-Year Consistency					
Metric	1-Year Windows	2-Year Windows	3-Year Windows	5-Year Windows	Placebo
Mean Spearman	0.233	0.321	0.381	0.457	0.010
Panel C: Fixed Annual Blocks, Varying Lag					
Lag g	1 year	2 years	3 years	5 years	
Pooled Pearson r	0.308	0.275	0.270	0.257	
Pooled Spearman r	0.251	0.220	0.216	0.193	
Mean Spearman	0.233	0.218	0.207	0.198	
N (trajectories)	1,295,590	1,272,410	1,248,355	1,200,545	

Table 6: PDR Quintile Transition Matrices

This table presents transition matrices. Each cell is the probability that a trajectory in in-sample quintile q_1 (row) falls in out-of-sample quintile q_2 (column). Quintiles are formed within each formation year; Q1 has the lowest PDR (most recurrent) and Q5 the highest. Higher diagonal values indicate stronger persistence. Panels A–D use the one-, two-, three-, and five-year comparison windows of Table 5, and Panel E shows the random-walk placebo with one-year windows. All analyses use data from 1971–2025.

Panel A: 1-Year					
	Q1	Q2	Q3	Q4	Q5
Q1	0.267	0.232	0.206	0.179	0.117
Q2	0.229	0.226	0.216	0.196	0.132
Q3	0.203	0.214	0.214	0.211	0.157
Q4	0.179	0.193	0.208	0.221	0.199
Q5	0.122	0.135	0.155	0.193	0.395
Panel B: 2-Year					
	Q1	Q2	Q3	Q4	Q5
Q1	0.295	0.242	0.205	0.165	0.093
Q2	0.241	0.238	0.221	0.190	0.109
Q3	0.205	0.219	0.224	0.216	0.136
Q4	0.165	0.192	0.214	0.237	0.191
Q5	0.095	0.108	0.135	0.191	0.471
Panel C: 3-Year					
	Q1	Q2	Q3	Q4	Q5
Q1	0.318	0.248	0.204	0.155	0.076
Q2	0.248	0.245	0.225	0.189	0.094
Q3	0.202	0.225	0.229	0.219	0.124
Q4	0.155	0.188	0.218	0.249	0.190
Q5	0.078	0.094	0.125	0.187	0.517
Panel D: 5-Year					
	Q1	Q2	Q3	Q4	Q5
Q1	0.348	0.258	0.198	0.139	0.056
Q2	0.256	0.255	0.231	0.183	0.075
Q3	0.199	0.229	0.239	0.224	0.109
Q4	0.142	0.181	0.223	0.266	0.188
Q5	0.055	0.077	0.109	0.188	0.571
Panel E: Placebo					
	Q1	Q2	Q3	Q4	Q5
Q1	0.226	0.186	0.183	0.185	0.219
Q2	0.188	0.209	0.209	0.208	0.186
Q3	0.183	0.211	0.213	0.209	0.183
Q4	0.186	0.207	0.211	0.209	0.186
Q5	0.217	0.186	0.183	0.189	0.225

Table 7: Sign Preservation Rate by Neighbor-Outcome Score and PDR Quintile

This table presents sign preservation rates for different window lengths. Sign preservation is the share of trajectories whose neighbor-outcome score has the same sign in-sample and out-of-sample. Panel A sorts all trajectories into four bins of the in-sample neighbor-outcome score $|T|$, with cutoffs at 1, 1.96, and 2.58. Panel B sorts all trajectories into Pattern Distance Ratio (PDR) quintiles and then keeps those with high scores ($|T| \geq 1.96$). Online Appendix B explains how to interpret the score. Q1 contains trajectories with the lowest PDR (highest recurrence), and Q5 contains trajectories with the highest PDR (lowest recurrence). Quintiles are formed within each formation year. The "Adj. Null" (independence-adjusted null) rows report the rate expected when a trajectory's in-sample and out-of-sample signs are independent: $P(\text{in}>0) \times P(\text{out}>0) + P(\text{in}<0) \times P(\text{out}<0)$, where the in-sample probabilities are measured within each bin and $P(\text{out}>0)$ is the unconditional probability from the full sample. The benchmark therefore differs from bin to bin. The comparison windows are one, two, three, and five years long, as in Table 5. The sample period is 1971–2025.

Panel A: All Trajectories by Neighbor-Outcome Score

$\|T\|$	1-Year	2-Year	3-Year	5-Year
$[0, 1)$	50.65%	50.40%	50.45%	50.50%
$[1, 1.96)$	51.51%	51.52%	51.47%	51.45%
$[1.96, 2.58)$	52.63%	52.69%	52.26%	52.62%
≥ 2.58	52.75%	53.33%	52.78%	53.81%
Adj. Null				
$[0, 1)$	50.00%	50.00%	49.99%	49.99%
$[1, 1.96)$	50.05%	50.05%	50.05%	50.05%
$[1.96, 2.58)$	50.19%	50.21%	50.23%	50.26%
≥ 2.58	50.33%	50.37%	50.48%	50.55%

Panel B: High Neighbor-Outcome Score ($|T| \geq 1.96$) by PDR Quintile

Quintile	1-Year	2-Year	3-Year	5-Year
Q1 (Low PDR)	52.89%	53.30%	52.34%	52.30%
Q2	53.01%	53.44%	52.30%	53.69%
Q3	52.74%	52.40%	52.64%	53.64%
Q4	52.64%	52.68%	52.99%	52.99%
Q5 (High PDR)	51.84%	52.40%	51.89%	52.43%
Adj. Null	50.24%	50.26%	50.31%	50.34%

Table 8: Portfolio Returns by Signal Quintile

This table reports annualized returns (%) of the five signal quintiles and of the long–short portfolio (Q5 − Q1) of Section 5.1 over the out-of-sample period 1975–2025. Q1 holds the stocks with the lowest signal and Q5 those with the highest. *Excess return* is the mean return in excess of the one-month Treasury bill rate (for Q5 − Q1, the mean long–short return), and *FF6* α is the alpha against the five Fama–French factors and momentum. Newey–West (6 lags) t-statistics in parentheses. ***, **, * denote significance at the 1%, 5%, and 10% levels.

	Equal-weighted		Value-weighted	
Quintile	Excess return	FF6 α	Excess return	FF6 α
Q1 (Low)	3.77 (1.31)	-5.74^{***} (-5.07)	3.94 (1.56)	-4.99^{***} (-3.99)
Q2	6.61^{**} (2.35)	-3.19^{***} (-2.83)	6.35^{***} (2.61)	-2.91^{***} (-2.95)
Q3	8.34^{***} (2.96)	-1.65 (-1.45)	7.51^{***} (3.19)	-1.61^{*} (-1.90)
Q4	9.30^{***} (3.32)	-0.73 (-0.63)	8.32^{***} (3.45)	-0.91 (-0.89)
Q5 (High)	12.53^{***} (4.55)	2.27^{*} (1.88)	10.44^{***} (4.71)	1.17 (1.10)
Q5 − Q1	8.76^{***} (9.46)	8.01^{***} (7.84)	6.50^{***} (5.17)	6.16^{***} (4.00)

Table 9: Long–Short Alphas: Factor Models and Subsample Stability

This table reports annualized alphas (%) of the long–short portfolio (Q5 − Q1) of Section 5.1. Portfolios are equal-weighted or value-weighted. Each panel reports alphas from seven models: the CAPM, the Fama–French three- and five-factor models (FF3, FF5) (Fama and French, 1993, 2015), the Carhart four-factor model (FF4) (Carhart, 1997), FF5 plus momentum (FF6), FF6 plus French's monthly short-term reversal factor (ST-Rev), and FF6 plus both ST-Rev and a weekly reversal factor (WST-Rev). We build WST-Rev in our universe. Each month, we sort trajectories on the compound return over the last five days of the formation window, buy the bottom quintile, sell the top quintile, and hold for twenty days. The factor is equal-weighted in the equal-weighted column and value-weighted in the value-weighted column. Panel A covers the out-of-sample period 1975–2025, and Panels B and C its most recent 26 and 17 years, 2000–2025 and 2009–2025. Newey–West (6 lags) t-statistics in parentheses. ***, **, * denote significance at the 1%, 5%, and 10% levels.

Panel A: Factor models, full sample (1975–2025)		
Model	Equal-weighted	Value-weighted
CAPM	9.20*** (9.41)	7.07*** (5.31)
FF3	9.08*** (9.10)	7.00*** (5.04)
FF4	8.40*** (8.32)	6.54*** (4.27)
FF5	8.50*** (7.92)	6.47*** (4.46)
FF6	8.01*** (7.84)	6.16*** (4.00)
FF6 + ST-Rev	8.21*** (7.74)	6.18*** (3.93)
FF6 + ST-Rev + WST-Rev	6.63*** (6.01)	5.98*** (3.69)
Panel B: Factor models, 2000–2025		
Model	Equal-weighted	Value-weighted
CAPM	9.33*** (6.26)	9.22*** (4.78)
FF3	9.10*** (6.29)	8.96*** (4.62)
FF4	8.71*** (5.88)	8.80*** (4.12)
FF5	8.30*** (5.81)	8.30*** (4.19)
FF6	8.07*** (5.68)	8.25*** (3.89)
FF6 + ST-Rev	8.11*** (5.77)	8.24*** (3.89)
FF6 + ST-Rev + WST-Rev	7.29*** (5.28)	8.31*** (3.88)
Panel C: Factor models, 2009–2025		
Model	Equal-weighted	Value-weighted
CAPM	7.56*** (4.33)	7.78*** (3.90)
FF3	7.53*** (4.47)	7.74*** (3.90)
FF4	7.34*** (4.34)	7.35*** (3.76)
FF5	7.32*** (4.36)	7.31*** (3.57)
FF6	7.16*** (4.20)	6.96*** (3.41)
FF6 + ST-Rev	7.11*** (4.27)	6.84*** (3.52)
FF6 + ST-Rev + WST-Rev	7.24*** (4.25)	7.24*** (3.67)

Table 10: Is the Signal Spanned by Technical Rules and Standard Predictors?

This table reports monthly Fama–MacBeth regressions of the next 20-day return, in percent, on the signal and other predictors, which Table A10 in Online Appendix D defines. *Signal* is the within-month percentile rank of the engine's prediction after the step-4 orthogonalization of Section 5.1. The trend forecast (Han et al., 2016) and the price delay (Hou and Moskowitz, 2005) come from Chen and Zimmermann (2022) and end in 2024. All regressors are winsorized at the 1st and 99th percentiles and standardized within the month. The sample runs from January 1975 to November 2025 (611 months, 1,705 stocks per month), and column (4) ends in July 2024 (595 months, 1,636 stocks per month). The regressions are equal-weighted, and Table A8 reports the value-weighted version. Newey–West (6 lags) t-statistics in parentheses. ***, **, * denote significance at the 1%, 5%, and 10% levels.

Regressor	(1)	(2)	(3)	(4)
Signal	0.185*** (7.84)	0.138*** (8.37)	0.169*** (9.18)	0.142*** (7.70)
MA ratio	−1.074*** (−7.81)	−1.032*** (−9.28)	−0.412*** (−4.13)	−0.347*** (−3.27)
MA crossover	0.898*** (7.03)	0.900*** (9.25)	0.309*** (3.45)	0.217** (2.20)
Reversal (21-day)		−0.275*** (−5.51)	−0.157*** (−3.43)	−0.114** (−2.46)
Momentum (12–2)		0.252*** (4.56)	0.242*** (4.46)	0.267*** (5.54)
Long-term reversal (60–25)		−0.032 (−1.15)	−0.034 (−1.22)	−0.039 (−1.41)
IVOL		−0.249*** (−4.96)	−0.252*** (−5.06)	−0.293*** (−6.02)
Max		0.024 (0.89)	0.033 (1.26)	0.062*** (2.61)
Beta		0.063 (0.95)	0.066 (1.06)	0.057 (0.85)
log market equity		−0.407*** (−3.68)	−0.423*** (−3.83)	−0.318*** (−3.42)
log book-to-market		0.039 (0.92)	0.035 (0.82)	0.022 (0.51)
Illiquidity (Amihud)		0.057** (2.45)	0.060*** (2.60)	0.063*** (3.01)
Turnover		−0.100* (−1.88)	−0.107** (−2.06)	−0.097* (−1.92)
Bid–ask spread		0.047 (0.98)	0.035 (0.74)	0.024 (0.56)
Dollar volume (log)		0.292*** (2.63)	0.308*** (2.80)	0.216** (2.43)
Trend forecast				0.087** (2.03)
Price delay				−0.011 (−0.64)
52-week high		0.086 (1.54)	0.058 (1.06)	−0.013 (−0.28)
Reversal (5-day)			−0.306*** (−7.42)	−0.259*** (−6.82)
Avg. adj. R^2 (%)	2.43	9.99	10.22	10.81

Table 11: Implementability: Turnover and Break-Even Trading Costs

This table reports the strategy's turnover and the trading cost that would use up its alpha. Panel A is the baseline monthly-rebalanced strategy of Table 9. Panel B restricts both legs to the largest NYSE size tercile of Table 12. EW and VW rows are equal- and value-weighted. *Gross* is the annualized FF6 alpha (%) with its Newey–West (6 lags) t-statistic in parentheses. *Turnover* is one-way turnover per leg, in percent per month. The break-even columns report, in basis points, a uniform one-way cost per dollar traded. *Raw* is the cost at which the raw long–short return is zero. It equals the average monthly long–short return divided by the dollars traded per month per dollar held on a side, counting every purchase and sale on both legs. The $\alpha = 0$ column is the cost at which the net FF6 alpha is zero. It equals the gross FF6 alpha, expressed per month, divided by the same dollars traded. The $t = 1.96$ column is the cost at which the t-statistic of the net alpha falls to 1.96. The round-trip cost is twice the figure shown. ***, **, * denote significance at the 1%, 5%, and 10% levels.

Panel A: Monthly rebalancing, all stocks

Sample		Gross FF6 α (%/yr)	Turnover (%/mo)	Break-even one-way cost (bp)		
				Raw	$\alpha = 0$	$t = 1.96$
Full (1975–2025)	EW	8.01*** (7.84)	82	22.2	20.3	15.1
	VW	6.16*** (4.00)	84	16.1	15.2	7.7
2000–2025	EW	8.07*** (5.68)	82	21.4	20.5	13.4
	VW	8.25*** (3.89)	84	20.2	20.4	10.1
2009–2025	EW	7.16*** (4.20)	82	17.4	18.1	9.6
	VW	6.96*** (3.41)	84	16.5	17.3	7.3

Panel B: Monthly rebalancing, largest NYSE size tercile

Sample		Gross FF6 α (%/yr)	Turnover (%/mo)	Raw	$\alpha = 0$	$t = 1.96$
Full (1975–2025)	EW	5.45*** (3.84)	82	15.0	13.9	6.8
	VW	5.17*** (3.14)	83	14.0	13.0	4.9
2000–2025	EW	6.54*** (3.20)	82	17.0	16.6	6.4
	VW	8.04*** (3.42)	83	20.1	20.3	8.7
2009–2025	EW	5.23** (2.41)	82	14.0	13.3	2.5
	VW	6.22*** (2.92)	82	15.8	15.7	5.2

Table 12: Signal Quality by Firm Size

This table reports the quality of the signal within firm-size terciles. The tercile breakpoints are the 33rd and 67th percentiles of NYSE market equity each month, applied to all stocks. The rank information coefficient (*rank-IC*) is the monthly unweighted Spearman correlation between the signal and the subsequent raw 20-day return across all stocks in the tercile. *Sharpe* is the annualized Sharpe ratio of the long–short portfolio formed inside the tercile, equal-weighted (EW) and value-weighted (VW). Results cover the full out-of-sample period (1975–2025) and its 2000–2025 and 2009–2025 subsamples. Newey–West (6 lags) t-statistics of the mean monthly rank-IC in parentheses; the significance markers apply to the rank-IC rows. ***, **, * denote significance at the 1%, 5%, and 10% levels.

	Small	Mid	Large
Full (1975–2025)			
rank-IC	0.026^{***} (11.3)	0.021^{***} (6.9)	0.020^{***} (5.1)
Sharpe (EW/VW)	1.40/1.23	0.96/0.93	0.71/0.62
2000–2025			
rank-IC	0.021^{***} (5.6)	0.020^{***} (4.2)	0.020^{***} (3.2)
Sharpe (EW/VW)	1.07/0.91	0.77/0.77	0.70/0.79
2009–2025			
rank-IC	0.018^{***} (3.8)	0.017^{***} (2.8)	0.022^{***} (2.9)
Sharpe (EW/VW)	0.82/0.65	0.69/0.70	0.66/0.75

Table 13: International Samples and PDR Persistence across Markets

This table describes the twelve international samples and reports PDR persistence in each market. The data and screens are described in Section 6.1. *Trajectories* is the number of 60-trading-day return trajectories ending at month-ends with complete data. *Engine OOS* is the out-of-sample prediction window of the engine in Section 6.2, and *Stocks/month* is the average number of scored stocks per month within it. The PDR follows the U.S. construction, with the three-shuffle random-walk benchmark (Section 2.3). Persistence is the cross-sectional correlation between in-sample PDR and one-year-ahead out-of-sample PDR in each market-year. We retain market-years in which at least 5,000 trajectories have both an in-sample and an out-of-sample PDR and the following year, the out-of-sample block, also has at least 5,000 trajectories. *Years* counts them, and *N* is their pooled number of trajectories. *Mean r* and *t* are the time-series mean and cross-year *t*-statistic of the annual Pearson correlations. *% Pos* is the share of market-years with a positive correlation. *Pooled r* and *Pooled* ρ are the Pearson and Spearman correlations computed over the pooled cross-section of all retained market-years. ***, **, * denote significance at the 1%, 5%, and 10% levels.

	Sample construction			PDR persistence						
Market	Trajectories	Stocks/ month	Engine OOS	Years	*N*	Mean *r*	*t*	% Pos	Pooled *r*	Pooled ρ
Hong Kong	209,142	669	2003–2025	13	137,689	0.507***	11.38	100.0	0.549	0.490
United Kingdom	249,102	692	1999–2025	29	231,243	0.500***	35.25	100.0	0.493	0.403
Malaysia	122,738	349	2002–2025	5	29,853	0.480***	9.04	100.0	0.463	0.407
Thailand	124,276	462	2008–2025	10	67,387	0.446***	11.01	100.0	0.456	0.390
Japan	967,615	2,341	1993–2025	36	952,840	0.361***	11.05	91.7	0.460	0.330
France	156,285	424	2000–2025	18	96,644	0.328***	14.18	100.0	0.318	0.261
Australia	128,415	400	2004–2025	5	30,583	0.318***	7.62	100.0	0.306	0.207
South Korea	492,962	1,529	2001–2025	26	464,207	0.292***	9.08	96.2	0.413	0.324
Sweden	93,338	340	2009–2025	5	29,193	0.227***	10.29	100.0	0.218	0.182
India	386,948	1,402	2005–2025	21	353,356	0.176***	7.12	95.2	0.198	0.186
Taiwan	401,613	1,151	1999–2025	24	359,546	0.146***	6.00	87.5	0.146	0.145
China	586,052	2,225	2005–2025	19	538,182	0.092***	3.12	73.7	0.143	0.128
Total	3,918,486	—	—	—	—	—	—	—	—	—

Table 14: The Engine in International Markets: Long–Short Alphas

This table reports **annualized** long–short (Q5 − Q1) alphas (%) of the engine of Section 6.2 in the twelve international markets of Table 13, equal- and value-weighted, for the full screened universe. Each market is benchmarked against its own regional five-factor model (Fama and French, 2017) augmented with the regional momentum factor of Fama and French (2012). Markets are grouped into French's Developed and Emerging universes and ordered within each group by Stage I PDR persistence. *Global* is the country-average portfolio of Chui et al. (2010), reported as a raw return. *Developed* and *Emerging* form the same country average within each universe, in months with at least three of its six markets active. Each is an alpha against French's Developed or Emerging five factors plus momentum, respectively. The t-statistics are Newey–West with 6 lags. ***, **, * denote significance at the 1%, 5%, and 10% levels.

Market	Months	EW	VW
Developed markets			
Hong Kong	276	10.54*** (5.93)	9.24*** (3.52)
United Kingdom	324	2.78* (1.86)	4.30 (1.48)
Japan	396	5.27*** (5.19)	5.34*** (3.66)
France	312	5.34*** (3.08)	2.04 (0.74)
Australia	264	11.26*** (4.34)	1.82 (0.67)
Sweden	204	13.81*** (5.75)	4.24 (1.19)
Emerging markets			
Malaysia	288	9.42*** (5.17)	12.06*** (3.71)
Thailand	216	2.34 (1.18)	12.22*** (4.14)
South Korea	300	4.12** (2.27)	−0.18 (−0.06)
India	252	12.22*** (7.13)	2.60 (0.77)
Taiwan	324	0.54 (0.23)	0.29 (0.08)
China	245	4.29* (1.77)	−0.31 (−0.08)
Global, raw return	300	6.79*** (11.77)	4.66*** (5.11)
Developed, factor alpha	312	8.48*** (9.83)	6.01*** (3.75)
Emerging, factor alpha	288	6.36*** (7.44)	4.37*** (2.99)

Table 15: The Engine in International Markets: Fama–MacBeth Regressions

This table reports, for the engine of Section 6.2 in the twelve international markets of Table 13, pooled cross-country Fama–MacBeth regressions of the 20-day forward return, demeaned within each country-month, on within-country-month percentile ranks of the signal and the controls. The coefficients are in percent per holding period, not annualized. When book equity is missing or non-positive, when December market equity is missing, or when the two are reported in different currencies (25% of firm-months), the log book-to-market ratio is set to zero and an availability indicator (*B/M available*) is added (Pontiff and Woodgate, 2008; McLean et al., 2009). The last column adds Amihud (2002) illiquidity. The t-statistics are Newey–West with 6 lags, from 396 monthly slopes (395 in the last column). ***, **, * denote significance at the 1%, 5%, and 10% levels.

	Univariate	Multivariate	+ Weekly rev.	+ Illiquidity
Signal	0.59^{***} (9.76)	0.54^{***} (9.62)	0.55^{***} (9.73)	0.55^{***} (9.81)
log market equity		-0.84^{***} (-3.91)	-0.80^{***} (-3.72)	-0.48 (-1.45)
log book-to-market		0.66^{***} (2.91)	0.66^{***} (2.95)	0.63^{***} (2.90)
B/M available		0.40^{***} (3.34)	0.40^{***} (3.34)	0.39^{***} (3.28)
Reversal (monthly)		-0.61^{***} (-2.82)	-0.20 (-1.06)	-0.17 (-0.92)
Reversal (weekly)			-0.99^{***} (-6.25)	-1.01^{***} (-6.79)
Momentum (12–2)		0.54^{*} (1.83)	0.57^{**} (1.98)	0.61^{**} (2.13)
Illiquidity (Amihud)				0.45^{*} (1.65)

Online Appendix

A. Summary of Variables and Notation

Table A1: Summary of Variables and Notation

This table lists the notation of the paper and where each object is defined. Stocks are indexed by i or j, trading days by t or s, and $r_{i,t}$ is the daily return of stock i on day t. A return trajectory $n = (i, t)$ is the 60-day return series of stock i that ends on month-end day t; another trajectory is written n'. Nearest neighbors are indexed by $k = 1, \ldots, K$, calendar years by y, random-walk series by $m = 1, \ldots, 4$, Stage II test years by Y, and portfolio months by τ.

Symbol	Definition	Defined in
Sample and benchmark portfolios		
$ME_{j,t}$	Market equity of stock j on day t: $\lvert P_{j,t}\rvert \times SHROUT_{j,t}$, summed over share classes (CRSP price and shares outstanding)	§2.1
BE	Book equity: $SEQ + TXDITC - PS$ (Compustat stockholders' equity, deferred taxes and investment tax credit, and preferred stock)	§2.1
BM	Book-to-market ratio for the sorts of year y: $BE_{y-1}/ME_{Dec,\,y-1}$	§2.1
$p(j)$	Fama–French 25 size and book-to-market portfolio to which stock j is assigned	§2.1
Trajectories and distances		
$\mathbf{R}_{i,t}$	Trajectory $n = (i, t)$: the daily returns $(r_{i,t-59}, \ldots, r_{i,t})$ of stock i over the 60-day formation window ending on day t	(1)
$d_{n',n}$	Z-normalized Euclidean distance between trajectories n' and n, the paper's "L2 distance"	(2)
$\mathcal{D}_y$	Set of trajectories whose formation window ends in calendar year y: the annual comparison block	§2.4
K	Number of nearest neighbors, $K = 100$	§2.4
ν_k^{real}, $\nu_k^{rand,m}$	k-th nearest neighbor of trajectory n among the real trajectories, and among the trajectories of random-walk series m	(4)–(5)
Random-walk benchmark and PDR		
$\mathcal{R}_y$, N_y	Pool of all daily returns of all stocks in year y, and its size; the random-walk series are drawn from it	(3)
$\bar{d}_n^{real\to real}$	Mean distance from trajectory n to its K nearest real neighbors	(4)

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Table A1 – continued from previous page

Symbol	Definition	Defined in
$\bar{d}_n^{real \to rand}$	Mean distance from trajectory n to its K nearest neighbors in each of random-walk series 1–3, averaged over the three series	(5)
PDR_n	Pattern Distance Ratio: $\bar{d}_n^{real \to real} / \bar{d}_n^{real \to rand}$; values below one indicate recurrence	(6)
PDR_n^{in}, $PDR_n^{out,h}$	PDR of trajectory n measured against the trajectories of years $y - h + 1$ to y, and of years $y + 1$ to $y + h$	§2.4
h	Comparison-window length in years (1, 2, 3, or 5)	§2.4
g	Lag in years between one-year blocks (Table 5, Panel C)	§2.4
$\mathbf{P}^h$, $P^h_{q_1,q_2}$	PDR quintile transition matrix for window length h, and the probability of moving from in-sample quintile q_1 to out-of-sample quintile q_2	(7)

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Table A1 – continued from previous page

Symbol	Definition	Defined in
Neighbor-outcome score		
R_n^{FF25}	FF25-adjusted cumulative return of trajectory n over the 20-day evaluation window	(8)
$r_{p(j),t+s}$	Daily value-weighted return of benchmark portfolio $p(j)$ on day $t+s$	(8)
$\bar{R}_n^{FF25}$, $\hat{\sigma}_n^{FF25}$	Mean and standard deviation of R^{FF25} across the K nearest neighbors of trajectory n	(9)
T_n	Neighbor-outcome score: $\bar{R}_n^{FF25} \big/ (\hat{\sigma}_n^{FF25}/\sqrt{K})$	(9)
T_n^{in}, $T_n^{out,h}$	Score measured against the in-sample and the out-of-sample window	§4
Stage II portfolios and factors		
$\mathcal{S}_Y$	Landmark set for test year Y	App. B
$Q_{q,\tau}$, w_n	Trajectories assigned to quintile q of the Section 5.1 signal at the end of month $\tau-1$; market equity of trajectory n's share class at the start of its evaluation window	(10)–(11)
R_n^{raw}	Unadjusted cumulative return of trajectory n over the evaluation window	(10)
$R_{q,\tau}^{EW}$, $R_{q,\tau}^{VW}$	Equal- and value-weighted return of quintile q in month τ	(10)–(11)
$R_{LS,\tau}$	Return of the long–short portfolio (Q5 minus Q1) formed at the end of month $\tau-1$, matched to the factors of month τ	App. B
$MktRf_\tau$, SMB_τ, HML_τ, RMW_τ, CMA_τ, UMD_τ	Market excess return and the size, value, profitability, investment, and momentum factors, from Kenneth French's data library	App. B
$STRev_\tau$, $WSTRev_\tau$	French's monthly short-term reversal factor, and the weekly reversal factor built in our universe under the strategy's own portfolio rule	App. B

B. Method Details

This appendix provides the formal definitions of the return measures, the block-bootstrap surrogate benchmark of Section 3.3, the landmark filters that define the landmark set, and the portfolio-formation and factor-model conventions used in Section 5. Table A9 in Online Appendix D lists the parameters of both stages.

Return Measures

For each return trajectory n associated with stock j, we compute the Fama–French 25 size-BM characteristic-adjusted return (Fama and French, 1993) over the 20-day *evaluation window* immediately following the *formation window*:

$$R_n^{FF25} = \prod_{s=1}^{20} \left(1 + r_{j,t+s}\right) - \prod_{s=1}^{20} \left(1 + r_{p(j),t+s}\right), \tag{8}$$

where t is the last day of the *formation window* and $r_{p(j),t+s}$ is the daily value-weighted return of the FF25 benchmark portfolio to which stock j is assigned. We evaluate predictability by examining the future FF25-adjusted returns of each trajectory's $K = 100$ nearest neighbors. The neighbor-outcome score is:

$$T_n = \frac{\bar{R}_n^{FF25}}{\hat{\sigma}_n^{FF25}/\sqrt{K}}, \tag{9}$$

where $\bar{R}_n^{FF25}$ and $\hat{\sigma}_n^{FF25}$ are the sample mean and standard deviation of the neighbors' future FF25-adjusted returns. A large $|T_n|$ means that trajectories similar to n have been followed by abnormal returns in a consistent direction. These FF25-adjusted returns are the main outcome variable. We use them to compute the scores above, and, after the reversal residualization and rank transform of Section 5, they are also the prediction target for the engine.

We call T_n the *neighbor-outcome score* and interpret it throughout the paper as a signal-to-noise score that ranks trajectories, not as a test statistic. The neighbors' future returns are not independent observations. Some neighbors share window end dates, and in that case their 20-day evaluation windows overlap entirely. Neighbors may also load on common shocks beyond the

FF25 adjustment, and they can include the same stock at nearby dates. The $\sqrt{K}$ scaling therefore overstates precision, and the cutoff $|T_n| \geq 1.96$ does not carry a nominal five-percent size. We use 1.96 only as a conventional cutoff on this score.

The Block-Bootstrap Surrogate Benchmark

The surrogate benchmark in Table A2 replaces the shuffle benchmark with a different null. This null keeps each stock's estimated factor exposures and the realized factor returns. It also keeps each stock's residual distribution and the short-range dependence of its residuals. Residuals far apart in time usually come from different, randomly chosen parts of the stock's history. We build it in four steps.

Factor exposures. For each stock i and each calendar year y we regress the stock's daily excess return on the six Fama–French factors (FF5 plus UMD) and a constant, using that year's daily observations: $r_{i,t} - r_{f,t} = a_{i,y} + b'_{i,y} F_t + \varepsilon_{i,t}$, where F_t holds the six factor returns on day t. We estimate exposures year by year rather than over the full sample. This lets a stock's factor loadings drift over time. We require more than 200 trading days in the stock-year. Stock-years below that threshold have no estimated beta, so they contribute no surrogate values.

Idiosyncratic series. The idiosyncratic return is $e_{i,t} = r_{i,t} - \hat{b}'_{i,y} F_t$, the raw return net of the fitted factor component only. We leave in the intercept and the risk-free rate, because every 60-day trajectory is z-normalized before any distance is computed. That step removes any constant level.

Resampling. We resample each stock's $e_{i,t}$ independently, using the stationary bootstrap of Politis and Romano (1994). Block lengths are geometric with success probability $p = 1/L$, so the mean block length is L trading days. Resampling is circular. We draw blocks within each maximal run of consecutive trading days on which the stock is in the sample and has an estimated beta. We then write the resampled values back to the same calendar dates. Two days in the same run that are k trading days apart fall in the same block with probability $(1 - 1/L)^k$. Because each stock is resampled on its own, the surrogate removes the comovement that the six factors miss, such as common industry shocks. The shuffle benchmark removes this comovement as well, so the two benchmarks do not differ in this respect. The rule that compared trajectories never share

a formation window (Section 2.4) limits how much such comovement can make real trajectories look alike, and the strict neighbor rule of Table A2 tests this channel directly.

Reassembly. The surrogate return is $\tilde{r}_{i,t} = \hat{b}'_{i,y} F_t + \tilde{e}_{i,t}$, where $\hat{b}_{i,y}$ is the estimated exposure and $\tilde{e}_{i,t}$ the resampled idiosyncratic return on day t. We add the factor component back, and we keep the realized factor series F_t in its original time order, attached to its own date. Every surrogate stock therefore carries the same real factor path as the data. By construction, comovement through the six factors has the same level and timing as in the real panel. We generate three independent replicates, the same number as the series of the random-walk benchmark. More replicates would give the denominator less sampling noise, so the two PDRs would differ for a reason other than the null, the only difference Panel A is meant to show. The ratio therefore keeps the form of Equation (6), $PDR_n = 3\,\bar{d}_n^{real\to real} / \sum_{m=1}^{3} \bar{d}_n^{real\to surr,m}$, where each denominator term is the mean distance to the $K = 100$ nearest neighbors within replicate m. Each replicate supplies one surrogate path for every stock-day that has an estimated beta. As a result, the benchmark pool is the same panel of stocks and dates as the real cross-section, minus the stock-years that the 200-day rule removes. Each real trajectory is still a query against this pool. Because the daily return sample begins in July 1971 (Section 2.1), no 1971 stock-year meets the 200-day rule, so the surrogate benchmark begins with formation year 1972 (Table A2).

The two nulls randomize at different granularities. For each replicate, the bootstrap runs once over the whole panel, not year by year, and it draws blocks along a stock's own history. The shuffle benchmark it replaces instead permutes returns across all stocks and trading days within a single year. The two nulls are aligned where the comparison happens, not where the randomization happens. The surrogate panel is cut into the same annual blocks $\mathcal{D}_y$, carries the same trajectory identifiers as the real panel, and runs through the same distance and neighbor code. The surrogate PDR can therefore be ranked and correlated within each formation year exactly as the baseline PDR is, which is what Panels A and B of Table A2 use. The main specification sets $L = 10$ trading days, and we report $L = 20$ alongside it.

Landmark Selection

For each test year Y, we take the landmark set from the pool of trajectories in the trailing five years ($Y-5$ through $Y-1$). The selection is point-in-time: each pool trajectory's PDR and $|T_n|$ are measured against the trajectories of the three calendar years before the trajectory's own year, so every statistic uses only returns realized before year Y:

1. **Score filter ($|T_n| \geq 1.96$).** We keep trajectories whose neighbor-outcome score (Equation 9) satisfies $|T_n| \geq 1.96$. As a result, landmarks are placed only where the local region of trajectory space has shown a high score.

2. **Recurrence restriction (the most recurrent 20%).** We take the 20th percentile of PDR over all pool trajectories with a finite PDR, and we keep the retained trajectories at or below it. The grid thus lies in the recurrent region of trajectory space. Every retained trajectory is a landmark: there is no cap on their number and no floor below which a test year is skipped. Table A7 reports the counts, from 424 in 1975 to 4,780 in 2025, 2,792 per test year on average.

We denote the resulting landmark set by $\mathcal{S}_Y$. The similarity between trajectory n and landmark ℓ is $\exp(-d_{n,\ell}^2/\omega_Y^2)$, where $d_{n,\ell}$ is the distance of Equation (2) and ω_Y^2 is the median squared distance between pairs of landmarks in $\mathcal{S}_Y$. In solving the ridge, eigenvalues of the centered feature Gram matrix below 10^{-4} of the largest are set to zero, because at that scale the matrix is dominated by noise.

Portfolio Formation

On each month-end date in the test year, we sort all trajectories by the trading signal of Section 5.1 and assign them to five quintile portfolios, Q1 (lowest) through Q5 (highest). We construct two types of portfolios:

- **Equal-weighted (EW):** The portfolio return for quintile q in month τ is the simple average of the holding-period returns of all trajectories assigned to quintile q at the end of month

$\tau - 1$:

$$R_{q,\tau}^{EW} = \frac{1}{|Q_{q,\tau}|} \sum_{n \in Q_{q,\tau}} R_n^{raw}, \tag{10}$$

where $Q_{q,\tau}$ denotes this set of trajectories, and $R_n^{raw} = \prod_{s=1}^{20}(1 + r_{j,t+s}) - 1$ is the unadjusted cumulative holding-period return. We use *raw* (unadjusted) returns to evaluate portfolios because these are the actual returns an investor would earn. We use the FF25-adjusted returns to build the prediction target (Section 5), so that the model targets abnormal performance rather than systematic factor exposures.

- **Value-weighted (VW):** The portfolio return for quintile q in month τ is the weighted average of holding-period returns, where the weights are market capitalizations:

$$R_{q,\tau}^{VW} = \frac{\sum_{n \in Q_{q,\tau}} w_n \cdot R_n^{raw}}{\sum_{n \in Q_{q,\tau}} w_n}, \tag{11}$$

where w_n is the market equity of trajectory n's share class, its price times shares outstanding, at the beginning of the *evaluation window*. To mitigate the influence of extreme weights, we winsorize market capitalizations at the 1st and 99th percentiles within each date before computing the weighted average.

The **long–short portfolio** is Q5 minus Q1. It goes long the highest-signal quintile and short the lowest.

Factor Model Specifications

We evaluate the long–short portfolio's risk-adjusted performance by regressing its monthly returns on standard factor models. Let $R_{LS,\tau}$ denote the return of the long–short portfolio formed at the end of month $\tau - 1$, earned over the following 20 trading days and matched to the factor returns of month τ. We estimate alphas from the following seven models:

1. **CAPM:** $R_{LS,\tau} = \alpha + \beta_{mkt} \cdot MktRf_\tau + \varepsilon_\tau$

2. **Fama–French 3-factor (FF3):** $R_{LS,\tau} = \alpha + \beta_{mkt} \cdot MktRf_\tau + \beta_{smb} \cdot SMB_\tau + \beta_{hml} \cdot HML_\tau + \varepsilon_\tau$

3. **Carhart 4-factor (FF4):** FF3 + $\beta_{umd} \cdot UMD_\tau$

4. **Fama–French 5-factor (FF5):** FF3 + $\beta_{rmw} \cdot RMW_\tau + \beta_{cma} \cdot CMA_\tau$

5. **6-factor (FF6):** FF5 + $\beta_{umd} \cdot UMD_\tau$

6. **FF6 + ST-Rev:** FF6 + $\beta_{strev} \cdot STRev_\tau$, where $STRev$ is French's monthly short-term reversal factor

7. **FF6 + ST-Rev + WST-Rev:** FF6 + ST-Rev + $\beta_{wstrev} \cdot WSTRev_\tau$, where $WSTRev$ is the weekly reversal factor built inside our universe under the strategy's own portfolio rule (Table 9)

where $MktRf$ is the market excess return, SMB is the size factor, HML is the value factor, UMD is the momentum factor, RMW is the profitability factor, and CMA is the investment factor. All factor returns except WST-Rev come from Kenneth French's data library. We construct WST-Rev as described in item 7 above. The intercept α in each regression is the abnormal return that the corresponding set of risk factors cannot explain. All returns and alphas are annualized by multiplying monthly values by 12, and Sharpe ratios are annualized by multiplying the monthly Sharpe ratio by $\sqrt{12}$. Throughout the paper, t-statistics are Newey–West with 6 lags (Newey and West, 1987). The automatic rule of Newey and West (1994) sets the lag selection parameter to $4(M/100)^{2/9}$, which is 5.98 at the $M = 611$ monthly observations of the out-of-sample period; we round up to six and apply the same lag to every regression and every time-series mean. The one exception is Table 13, whose t is the cross-year t-statistic of a market's annual correlations, a series of five to thirty-six observations.

C. Robustness

This appendix collects the robustness evidence for Sections 3 and 4. The Stage II robustness checks are reported in Section 5.2. Tables A2 through A4 concern the recurrence measure itself. The recurrence measure rests on three choices: the random-walk benchmark, the nearest-neighbor rule, and the trajectory length. Table A2 replaces the benchmark with a stationary block bootstrap of factor-model residuals and replaces the neighbor rule with the strict neighbor rule. It then re-runs the downstream transition analysis against this harder null. Table A3 asks whether PDR persistence is a restatement of volatility persistence. Table A4 varies the trajectory length and the holding period. Table A5 varies the trajectory length and the holding period together in the sign-preservation test. Table A6 repeats the baseline cell of that test under the strict neighbor rule. Finally, two Stage II tables also appear here: Table A7 reports the size of the landmark set in each test year, and Table A8 reports the value-weighted version of the horse race in Table 10.

Table A2: Robustness: PDR Persistence and Transition Matrix under Alternative Benchmark and Neighbor Definitions

This table checks PDR persistence under two alternative constructions of the PDR. Panel A reports, for each construction, the mean Pearson and Spearman correlations between in-sample and out-of-sample PDR across the formation years of the 1-year windows. *Strict neighbors* selects the $K = 100$ nearest neighbors with at most one neighbor per cell, where a cell is a size half $\times$ a book-to-market half (both split at the NYSE median) $\times$ one of 48 industries $\times$ the formation month; a trajectory that lacks one of these assignments forms its own cell. *Block-bootstrap surrogate* replaces the shuffle benchmark with a stationary block bootstrap of each stock's six-factor (FF5+UMD) residuals, and L is its mean block length in trading days. Online Appendix B gives the full construction. The Baseline and Random-walk placebo rows use the same in-sample and out-of-sample PDRs as the 1-Year Windows and Placebo columns of Table 5. Panel B reports the 1-year quintile transition matrix of the PDR measured against the $L = 10$ surrogate, with quintiles formed within each formation year. Its counterpart under the shuffle benchmark is Panel A of Table 6. All results use 60-day trajectories and the 1971–2025 sample, except that the surrogate results in both panels start with formation year 1972.

Panel A: PDR persistence under alternative constructions

Construction	Mean Pearson r	Mean Spearman r
Baseline	0.317	0.233
Strict neighbors	0.350	0.250
Block-bootstrap surrogate ($L = 10$)	0.247	0.176
Block-bootstrap surrogate ($L = 20$)	0.262	0.184
Random-walk placebo	0.014	0.010

Panel B: 1-year transition matrix, surrogate-based PDR ($L = 10$)

	Q1	Q2	Q3	Q4	Q5
Q1	0.339	0.200	0.167	0.149	0.145
Q2	0.199	0.216	0.206	0.199	0.180
Q3	0.165	0.207	0.215	0.214	0.200
Q4	0.150	0.198	0.212	0.220	0.219
Q5	0.147	0.179	0.200	0.218	0.257

Table A3: PDR Persistence and the Volatility Channel

The dependent variable is the one-year-ahead out-of-sample PDR, $PDR_n^{\text{out},1}$. The regressors are the trajectory's in-sample PDR and four controls measured on the same 60-day formation window. *IVOL* is the standard deviation of the residual from regressing the stock's daily return on the market return over those 60 days. *Log ME* is log market equity on the window's last day, *Reversal* is the compound return over its last 21 days, and *Trend* is the slope of the trajectory's standardized cumulative log-price path. Coefficients are Fama–MacBeth averages of the 54 cross-sectional regressions, one per formation year. Every variable is standardized within each cross-section. Newey–West (6 lags) t-statistics in parentheses. Avg. R^2 is the mean of the per-cross-section R^2. All three columns use one common sample. The lower panel forms IVOL terciles within each formation year. It reports the Pearson correlation between in-sample and out-of-sample PDR in each tercile, averaged across formation years. ***, **, * denote significance at the 1%, 5%, and 10% levels.

	Dependent variable: $PDR_n^{\text{out},1}$		
Regressor	(1)	(2)	(3)
In-sample PDR (PDR_n^{in})	0.317^{***} (6.48)	0.304^{***} (6.30)	0.304^{***} (6.29)
IVOL		0.093^{***} (7.61)	0.095^{***} (5.97)
Log ME			0.003 (0.22)
Reversal (monthly, 21-day)			0.017^{***} (3.54)
Trend			0.002 (0.20)
Avg. R^2	0.125	0.136	0.144
Observations	1,295,590	1,295,590	1,295,590
Mean Pearson persistence correlation within IVOL terciles			
Low IVOL	0.215 (positive in 49 of 54 formation years)		
Mid IVOL	0.270 (positive in 53 of 54 formation years)		
High IVOL	0.389 (positive in 54 of 54 formation years)		

Table A4: Robustness: PDR Persistence by Trajectory Length and Holding Period

This table reports PDR persistence over the grid of trajectory lengths $\{30, 60\}$ days and holding periods $\{20, 30\}$ days (columns 20d and 30d). The baseline is the 60-day trajectory with a 20-day holding period. The holding period does not enter the PDR. It changes only which trajectories qualify, because a trajectory needs complete data for its formation window plus the holding period. Panel A reports the pooled Pearson and Spearman correlations between in-sample and out-of-sample PDR, with N in thousands of trajectories. Panel B reports year-by-year consistency: the mean and standard deviation of the yearly Spearman correlations. All specifications use one-year comparison blocks with data from 1971–2025.

Panel A: Pooled Correlations

	30-day trajectory		60-day trajectory	
Metric	20d	30d	20d	30d
Pearson r	0.333	0.330	0.308	0.304
Spearman r	0.289	0.288	0.251	0.250
N (000s)	1,314	1,302	1,296	1,283

Panel B: Year-by-Year Consistency

	30-day trajectory		60-day trajectory	
Metric	20d	30d	20d	30d
Mean Spearman	0.274	0.273	0.233	0.232
Std Spearman	0.157	0.157	0.124	0.123

Table A5: Robustness: Sign Preservation Rate by Trajectory Length and Holding Period

This table reports sign preservation rates for the full grid of trajectory lengths $\{30, 60\}$ days and holding periods $\{20, 30\}$ days (columns 20d and 30d), where the holding period is the evaluation window over which the neighbor-outcome score of Equation (9) is measured. The baseline is the 60-day trajectory with a 20-day holding period. As in Table 7, Panel A sorts all trajectories into four bins of the in-sample neighbor-outcome score $|T|$, with cutoffs at 1, 1.96, and 2.58, and Panel B sorts all trajectories into PDR quintiles, formed within each formation year, and then keeps those with high scores ($|T| \geq 1.96$). Sign preservation is the share of trajectories whose neighbor-outcome score has the same sign in-sample and out-of-sample. The "Adj. Null" rows are the independence-adjusted null benchmark, computed as in Table 7. All specifications use one-year blocks with data from 1971–2025.

Panel A: All Trajectories by Neighbor-Outcome Score

	30-day trajectory		60-day trajectory	
$\|T\|$	20d	30d	20d	30d
$[0, 1)$	50.61%	50.42%	50.65%	50.36%
$[1, 1.96)$	51.60%	51.00%	51.51%	51.16%
$[1.96, 2.58)$	52.31%	51.50%	52.63%	51.90%
≥ 2.58	52.88%	51.51%	52.75%	51.96%
Adj. Null				
$[0, 1)$	50.00%	50.00%	50.00%	49.99%
$[1, 1.96)$	50.06%	50.03%	50.05%	50.03%
$[1.96, 2.58)$	50.18%	50.13%	50.19%	50.12%
≥ 2.58	50.30%	50.27%	50.33%	50.26%

Panel B: High Neighbor-Outcome Score ($|T| \geq 1.96$) by PDR Quintile

	30-day trajectory		60-day trajectory	
Quintile	20d	30d	20d	30d
Q1 (Low PDR)	53.26%	51.76%	52.89%	52.30%
Q2	52.51%	51.28%	53.01%	52.74%
Q3	52.72%	51.43%	52.74%	52.22%
Q4	52.13%	51.81%	52.64%	51.28%
Q5 (High PDR)	51.52%	51.11%	51.84%	50.66%
Adj. Null	50.22%	50.18%	50.24%	50.17%

Table A6: Robustness: Sign Preservation Rate under the Strict Neighbor Rule

This table repeats the baseline specification of Table A5 (60-day trajectories and a 20-day holding period) under the strict neighbor rule of Table A2, which keeps at most one neighbor per size × book-to-market × 48-industry × month cell. The Baseline column repeats the 60-day, 20d column of Table A5. Panels, quintiles, and the "Adj. Null" rows are as in Table A5. The strict rule leaves 105,857 high-score trajectories against 134,545 in the baseline. One-year blocks, 1971–2025.

Panel A: All Trajectories by Neighbor-Outcome Score		
$\lvert T\rvert$	Baseline	Strict neighbors
$[0, 1)$	50.65%	50.56%
$[1, 1.96)$	51.51%	51.76%
$[1.96, 2.58)$	52.63%	52.93%
≥ 2.58	52.75%	53.53%
Adj. Null		
$[0, 1)$	50.00%	50.00%
$[1, 1.96)$	50.05%	50.06%
$[1.96, 2.58)$	50.19%	50.21%
≥ 2.58	50.33%	50.40%
Panel B: High Neighbor-Outcome Score ($\lvert T\rvert \geq 1.96$) by PDR Quintile		
Quintile	Baseline	Strict neighbors
Q1 (Low PDR)	52.89%	53.83%
Q2	53.01%	53.44%
Q3	52.74%	53.40%
Q4	52.64%	52.32%
Q5 (High PDR)	51.84%	52.29%
Adj. Null	50.24%	50.26%

Table A7: The Landmark Set, Year by Year

This table reports the size of the landmark set behind every U.S. test year of Section 5.1. *Pool* is the number of 60-day trajectories formed in the five years before the test year. $|T| \geq 1.96$ counts those whose neighbor-outcome score clears the screen, and *Landmarks* those that also lie at or below the 20th percentile of the pool's PDR distribution; the third column is a subset of the second; because the 20th percentile is taken over the whole pool, the order of the two screens does not matter. Every such trajectory is a landmark, with no cap and no floor (Online Appendix B).

Year	Pool	$\lvert T\rvert \geq 1.96$	Landmarks	Year	Pool	$\lvert T\rvert \geq 1.96$	Landmarks
1975	54,270	1,612	424	2001	168,225	12,902	3,123
1976	74,484	3,239	802	2002	166,884	13,697	3,486
1977	90,676	4,859	1,175	2003	162,181	14,183	3,736
1978	97,890	6,318	1,505	2004	155,374	14,957	4,117
1979	102,703	7,846	1,816	2005	148,186	15,086	4,197
1980	104,327	7,669	1,686	2006	140,746	14,142	4,012
1981	105,635	7,729	1,712	2007	133,685	13,168	3,741
1982	106,573	8,110	1,811	2008	128,441	12,360	3,476
1983	106,969	8,435	1,885	2009	125,203	11,567	3,275
1984	107,830	8,562	1,960	2010	122,887	11,675	3,468
1985	108,960	8,668	1,970	2011	120,510	12,913	4,030
1986	110,086	8,593	1,947	2012	118,207	14,215	4,463
1987	111,561	8,100	1,848	2013	116,577	14,502	4,679
1988	112,807	7,756	1,726	2014	115,030	14,209	4,679
1989	113,256	7,658	1,725	2015	113,625	13,256	4,309
1990	113,410	7,724	1,774	2016	113,755	11,563	3,495
1991	113,882	7,773	1,814	2017	114,653	10,286	2,966
1992	114,696	8,022	1,916	2018	114,823	10,234	2,836
1993	117,092	8,421	2,111	2019	115,088	10,441	2,944
1994	122,032	8,878	2,216	2020	115,415	10,570	3,021
1995	129,068	9,499	2,288	2021	115,598	10,678	3,152
1996	137,624	9,914	2,299	2022	115,829	11,434	3,571
1997	147,105	10,513	2,368	2023	116,974	12,698	4,122
1998	156,056	11,065	2,429	2024	118,418	14,151	4,386
1999	162,744	11,416	2,498	2025	118,678	15,411	4,780
2000	166,764	11,724	2,647				

Table A8: The Horse Race, Value-Weighted

This table repeats Table 10 with value-weighted Fama–MacBeth regressions. We estimate each month's cross-sectional regression by weighted least squares, using market equity as the weight (Medhat and Schmeling, 2022; Green et al., 2017). The columns, the sample, the variable definitions, the winsorization and standardization, and the construction of the Newey–West (6 lags) t-statistics are exactly the same as in Table 10. Only the weighting differs. ***, **, * denote significance at the 1%, 5%, and 10% levels.

Regressor	(1)	(2)	(3)	(4)
Signal	0.138*** (3.86)	0.074*** (2.96)	0.090*** (3.49)	0.073*** (2.80)
MA ratio	−1.047*** (−5.53)	−1.023*** (−7.28)	−0.552*** (−3.45)	−0.529*** (−3.38)
MA crossover	0.982*** (5.23)	0.843*** (6.41)	0.397*** (2.81)	0.350** (2.40)
Reversal (21-day)		−0.094 (−1.39)	−0.013 (−0.20)	−0.003 (−0.04)
Momentum (12–2)		0.334*** (5.15)	0.323*** (5.07)	0.342*** (5.38)
Long-term reversal (60–25)		−0.072 (−1.64)	−0.073* (−1.70)	−0.070* (−1.68)
IVOL		−0.283*** (−4.08)	−0.275*** (−4.04)	−0.285*** (−4.40)
Max		0.058 (1.35)	0.058 (1.37)	0.059 (1.46)
Beta		0.077 (0.88)	0.091 (1.07)	0.049 (0.56)
log market equity		0.047 (0.33)	0.039 (0.28)	0.013 (0.10)
log book-to-market		0.011 (0.26)	0.007 (0.17)	−0.005 (−0.11)
Illiquidity (Amihud)		0.112** (2.16)	0.111** (2.14)	0.108** (2.10)
Turnover		0.106 (1.44)	0.104 (1.43)	0.105 (1.40)
Bid–ask spread		−0.003 (−0.04)	−0.024 (−0.37)	−0.001 (−0.01)
Dollar volume (log)		−0.112 (−0.83)	−0.109 (−0.81)	−0.084 (−0.62)
Trend forecast				0.022 (0.34)
Price delay				−0.008 (−0.27)
52-week high		−0.134* (−1.85)	−0.157** (−2.18)	−0.160** (−2.34)
Reversal (5-day)			−0.219*** (−4.61)	−0.208*** (−4.14)
Avg. adj. R^2 (%)	5.59	20.17	20.62	21.36

D. Parameters and Definitions

Table A9: Parameters of the Method

This table lists the parameters of both stages in the paper's configuration, for the United States and for the twelve international markets. *Same* means that the international value equals the U.S. value.

Parameter	United States	International
Trajectories and recurrence (Stage I)		
Trajectory	60 trading days ending on the last trading day of a month; returns required on all 60 days and on the 20 trading days that follow	Same
Normalization and distance	Z-normalized trajectories, Euclidean distance; trajectories that share a formation window are not compared	Same
Nearest neighbors K	100	Same
Random-walk benchmark	Three shuffles of all daily returns across all stocks and trading days within each calendar year; the placebo uses a further shuffle	Three shuffles; no placebo
PDR	Mean distance to the K nearest real trajectories over its mean across the three random-walk series	Same
Comparison windows	1, 2, 3, and 5 years	One year
Forward return	20 trading days after the formation date, in excess of the stock's FF25 value-weighted benchmark portfolio	20 trading days, in excess of the equal-weighted average of the market's other stocks
Neighbor-outcome score T	Mean of the K neighbors' forward returns over its standard error	Same
The engine (Stage II)		
Landmark pool	Trajectories of the five years before the test year	Same
Pool statistics	PDR and T of each pool trajectory, measured against the three preceding years	Same
Landmark rule	$\lvert T\rvert \geq 1.96$ and PDR at or below the pool's 20th percentile; no cap, no floor	Same
Features	Gaussian-kernel similarity $\exp(-d^2/\omega^2)$ to each landmark, centered	Same
Bandwidth ω^2	Median squared distance between the test year's landmarks	Same

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Table A9 – continued from previous page

Parameter	United States	International
Ridge penalty	Generalized cross-validation over $\lambda = c\,\bar{e}$, with $\bar{e}$ the mean eigenvalue and $c \in \{10^{-4}, 10^{-3.5}, \ldots, 10^{2}\}$; directions with eigenvalues below 10^{-4} of the largest are dropped	Same
Training sample	Expanding: every trajectory formed before December of the year before the test year	Same, from the market's first year
Re-estimation	Once per test year	Same
Minimum training sample	50,000 trajectories	20,000 trajectories
Target	Forward return residualized each month on 5-day and 21-day reversal (each winsorized at the 1st and 99th percentiles), then rank-Gaussian	Same
Signal	Prediction residualized each month on the same two characteristics; its within-month percentile rank	Same
Portfolios and evaluation		
Portfolios	Monthly quintiles of the signal in months with at least 50 scored stocks; long Q5, short Q1; held 20 trading days	Same
Value weights	The security's market equity on the formation date, winsorized at the 1st and 99th percentiles within the month	The firm's market equity over its issues at the month's last price, in local currency, winsorized
Factor models	CAPM, FF3, FF4, FF5, FF6, FF6 + ST-Rev, FF6 + ST-Rev + WST-Rev	Regional five factors plus momentum
t-statistics	Newey–West, 6 lags	Same
Out-of-sample period	Formation months January 1975 to November 2025	First test year 1993 (Japan) to 2009 (Sweden), through December 2025

Table A10: Definitions of the Predictors

This table defines the predictors of Tables 10 and A8. Month t is the formation month, whose last trading day ends the 60-day formation window. The liquidity measures (illiquidity, turnover, bid–ask spread, and dollar volume) require at least six of their twelve months. All predictors are winsorized at the 1st and 99th percentiles and standardized within the month.

Predictor	**Definition**	**Source**
MA ratio	Price on the formation date over its average over the 60-day formation window, with prices built from the trajectory's daily returns	Brock et al. (1992)
MA crossover	Average price over the last 5 days of the formation window over the average over its last 50 days	Brock et al. (1992)
Reversal (21-day)	Compound return over the last 21 trading days of the formation window	Jegadeesh (1990)
Momentum (12–2)	Cumulative return over calendar months $t-11$ to $t-1$, from CRSP monthly returns; all eleven months required	Jegadeesh and Titman (1993)
Long-term reversal (60–25)	Cumulative log return over months $t-59$ to $t-24$, from CRSP monthly returns; at least 24 of the 36 months required	De Bondt and Thaler (1985)
IVOL	Standard deviation of the residuals from a regression of the stock's daily returns on the market return over the formation window	Ang et al. (2006)
Max	Largest daily return over the last 21 trading days of the formation window	Bali et al. (2011)
Beta	Slope from a regression of daily excess returns on the market excess return over months $t-11$ to t; at least 120 days	—
log market equity	Log of price times shares outstanding on the formation date	Banz (1981)
log book-to-market	Log of the book equity of the fiscal year ending in calendar year $y-1$ over the firm's market equity at the end of December of $y-1$, used for formation months from July of y to June of $y+1$	Fama and French (1993)
Illiquidity (Amihud)	Mean over months $t-11$ to t of the monthly average of daily absolute return over dollar volume	Amihud (2002)
Turnover	Mean over months $t-11$ to t of the monthly average of daily share volume over shares outstanding	—

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Table A10 – continued from previous page

Predictor	Definition	Source
Bid–ask spread	Mean over months $t-11$ to t of the monthly effective spread, the average of the two-day Corwin–Schultz and Abdi–Ranaldo estimates, each pair of days counted in the month of its second day	Corwin and Schultz (2012); Abdi and Ranaldo (2017)
Dollar volume (log)	Log of one plus the mean over months $t-11$ to t of the monthly average daily dollar volume	—
Trend forecast	The trend signal for month t in the open-source data of Chen and Zimmermann (2022); ends in December 2024	Han et al. (2016)
Price delay	The price-delay measure for month t in Chen and Zimmermann (2022), updated each July; ends in July 2024	Hou and Moskowitz (2005)
52-week high	Price at the end of month t over the highest daily price of months $t-11$ to t, with prices adjusted for splits	George and Hwang (2004)
Reversal (5-day)	Compound return over the last 5 trading days of the formation window	Lehmann (1990)

E. The International Markets

Table A11: The International Markets

This table describes the twelve international markets of Section 6. A market's sample holds the common shares of the firms headquartered in it that trade on the exchanges listed: the exchange with the most of the market's firms, or the two with the most in China, France, India, Japan, South Korea, and Taiwan. Exchange names are Compustat Global's. Returns are in the quotation currency, which is the local currency for at least 99 percent of the daily observations in every market. *Regional factors* are the Fama–French factors against which the market's long–short portfolio is evaluated in Table 14.

Market	Exchanges	Currency	Regional factors
Developed markets			
Australia	ASX All Markets	AUD	Asia Pacific ex Japan
France	NYSE Euronext Paris; Deutsche Boerse AG	EUR (FRF before 1999)	Europe
Hong Kong	Hong Kong Exchanges and Clearing Ltd	HKD	Asia Pacific ex Japan
Japan	Tokyo Stock Exchange; Tokyo Stock Exchange JASDAQ	JPY	Japan
Sweden	NASDAQ OMX Nordic	SEK	Europe
United Kingdom	London Stock Exchange	GBP	Europe
Emerging markets			
China	Shenzhen Stock Exchange; Shanghai Stock Exchange	CNY	Emerging Markets
India	BSE Ltd; National Stock Exchange of India	INR	Emerging Markets
Malaysia	Bursa Malaysia	MYR	Emerging Markets
South Korea	Korea Exchange KOSDAQ; Korea Exchange Stock Market	KRW	Emerging Markets
Taiwan	GreTai Securities Market; Taiwan Stock Exchange	TWD	Emerging Markets
Thailand	Stock Exchange of Thailand	THB	Emerging Markets